

The Geometry of Risk: An Integrated Monitoring Framework for Multi-Asset Systemic Stress

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Structured Abstract

Purpose – This paper develops an integrated four-layer empirical monitoring framework for diagnosing multi-asset systemic stress, positioned within the empirical systemic-risk tradition of Kritzman et al. (2011) and Diebold and Yilmaz (2014) rather than as a structural equilibrium asset-pricing model. The objective is to provide institutional risk managers and macro analysts with a reproducible diagnostic architecture combining network topology, dynamic causality analysis, tail-risk quantification, and regime classification, applied concurrently with an active stress event. The paper does not aim to forecast returns, identify the stochastic discount factor, or test market efficiency.

Design/methodology/approach – Four methodological layers are integrated: network topology via Mantegna (1999) correlation distance, multidimensional scaling, Ward clustering, and minimum spanning trees; dynamic causality via pairwise Granger testing with Bonferroni correction, lead-lag cross-correlation, and joint F-tests in a nine-asset VAR; tail risk via Conditional Value-at-Risk and Longin and Solnik (2001) tail dependence; regime classification via the Kritzman et al. (2011) Absorption Ratio and a hidden Markov model with BIC selection and out-of-sample evaluation. Analysis covers nine global asset classes from April 2025 to April 2026 ($N = 261$), supplemented by a 14-year long-history sample ($N = 3,472$).

Findings – The Absorption Ratio 0.50 threshold (Kritzman et al., 2011) is breached in all four crisis windows tested (the Global Financial Crisis, COVID-19, the 2022–2023 Federal Reserve tightening cycle, and the current-study window), providing multi-episode threshold validation. Bonferroni correction reduces 15 uncorrected Granger relationships to 4 — all pointing to Japan Equity. The DCC-GARCH minimum spanning tree converges with the Granger evidence on Japan Equity centrality. The out-of-sample hidden Markov model, trained on 14 years of multi-crisis

data and applied without re-estimation, correctly identifies the April 2026 stress regime. Joint F-tests support Oil quasi-exogeneity in conditional mean while variance decomposition reveals volatility-channel transmission to equities and rates.

Research limitations/implications – The 261-observation current-study window is short for systemic-risk modelling; conclusions specific to this window are exploratory. The multi-crisis Absorption Ratio replication, long-history DCC-GARCH baseline, and out-of-sample hidden Markov model evaluation mitigate but do not eliminate single-episode concerns. Granger findings are subject to heteroskedasticity, non-linear dependence, and finite-sample size distortions; only Bonferroni-surviving results are treated as robust. The framework provides no equilibrium interpretation. Future research should pursue walk-forward hidden Markov model evaluation on larger samples, asymmetric and block DCC specifications, Student-t and copula-based tail dependence extensions, and transfer-entropy or non-linear Granger frameworks for non-linear predictability structures.

Practical implications – The framework offers institutional risk managers a reproducible monitoring architecture combining four methodologically independent diagnostics. Portfolio managers obtain a multi-method check on diversification assumptions during stress periods. The volume-weighted minimum spanning tree provides a capital-commitment-aware view of cross-asset structure that complements the price-correlation view. The out-of-sample hidden Markov model enables real-time regime classification using only historical data. Risk officers receive defensible directional findings (Bonferroni-corrected Granger) that survive the most stringent multiple-testing correction. Macro strategists gain a transparent map of which assets transmit shocks through mean predictability versus through volatility-channel effects on forecast-error variance.

Social implications – Open and reproducible systemic-risk monitoring tools support broader financial-stability objectives valued by regulators, central banks, and institutional fiduciaries who serve household savers, pension beneficiaries, and insurance policyholders. The framework's open-source code and fully bundled data pipeline lower the methodological barrier for smaller institutions, academic researchers in emerging markets, and graduate students who may lack access to proprietary risk platforms. By providing methodologically triangulated diagnostics rather than single-method signals, the framework helps reduce overconfidence in any one indicator — a recurring contributor to retail and institutional losses during prior systemic crises.

Originality/value – The paper's contribution lies in integration and methodological triangulation rather than in any single new method. Four established methodological streams are unified into a single reproducible framework applied concurrently with an active stress event. The volume-

weighted minimum spanning tree extension incorporates capital-commitment data that price-only correlation approaches miss. The out-of-sample hidden Markov model evaluation on a 14-year multi-crisis training history addresses the in-sample-only limitation of prior regime-detection frameworks. The dedicated econometric-limitations subsection and DCC-GARCH parametric benchmark allow readers to assess which findings are method-robust and which are method-specific.

Keywords: systemic risk monitoring, minimum spanning tree, absorption ratio, hidden Markov model, Granger causality, Bonferroni correction, dynamic conditional correlation, tail dependence, regime detection, multi-asset portfolio diagnostics

Article classification: Research paper

1. Introduction

The April 2026 global market stress episode illustrated a recurring challenge in applied risk management: the standard toolkit of rolling correlations and linear volatility metrics provides limited advance warning of the cross-asset synchronisation that characterises systemic stress events. This paper develops the Geometry of Risk — an integrated four-layer empirical monitoring framework — applied to nine globally representative asset classes over April 2025 to 24 April 2026.

Before describing the framework, it is essential to be explicit about its scope and objectives. The Geometry of Risk is a risk-monitoring and diagnostic tool, not an equilibrium asset-pricing model, not a structural contagion model derived from first principles, and not a forecasting system. It operates within the empirical systemic-risk monitoring tradition established by Kritzman et al. (2011) and Diebold and Yilmaz (2014), whose frameworks are similarly positioned as diagnostic instruments rather than as rational-equilibrium models. The statistical objects produced by the framework — correlation distances, minimum spanning trees, principal components, Granger predictability matrices, and HMM latent states — are descriptive characterisations of the empirical joint distribution of returns. They do not, and are not claimed to, imply specific economic transmission mechanisms, rational-equilibrium pricing relations, or arbitrage-free valuation structures.

The paper makes four specific contributions, each motivated by a methodological gap in the existing systemic-risk literature. First, it provides an integrated four-layer monitoring architecture combining network topology, dynamic causality analysis, tail risk quantification, and HMM regime classification into a single reproducible framework applied concurrently with an active stress event. Second, it introduces a volume-weighted minimum spanning tree analysis that incorporates capital commitment data and reveals network configurations differing from price-only correlation topology. Third, it implements BIC-based HMM model selection with out-of-sample evaluation on a long-history

training sample, addressing the in-sample-only limitation of prior regime-detection frameworks. Fourth, it incorporates an explicit econometric-limitations subsection and a Dynamic Conditional Correlation GARCH baseline (Engle, 2002), enabling readers to calibrate the epistemic weight attached to each finding and to compare topological characterisations against the standard parametric benchmark for time-varying correlation.

Four research questions structure the empirical analysis. RQ1: How does the empirical correlation topology of a multi-asset global portfolio organise during a stress event, and which descriptive risk groupings remain structurally distinct? RQ2: Which assets exhibit network centrality, peripherality, and directional lead-lag patterns that survive multiple-testing correction? RQ3: What HMM-identified regime structure characterises the empirical dynamics of the Systemic Absorption Ratio during the current episode, and does this structure replicate out-of-sample when calibrated on a 14-year multi-crisis training history? RQ4: How do the identified descriptive patterns translate into risk-monitoring implications, conditional on the acknowledged econometric limitations of the constituent methods?

The paper is organised as follows. Section 2 reviews the relevant literature and situates the paper within the rational-finance landscape. Section 3 describes the data and methodology, including a dedicated econometric-limitations subsection. Section 4 presents empirical findings, integrating the multi-crisis Absorption Ratio replication, the Bonferroni-corrected Granger analysis, the DCC-GARCH baseline, and the out-of-sample HMM evaluation. Section 5 discusses implications and limitations. Section 6 concludes.

2. Theoretical Context and Literature Review

2.1. Positioning Within Rational-Finance Theory

The canonical framework of rational asset pricing, formalised by Cochrane (2005) and Duffie (2001), derives asset prices as expected discounted payoffs under a Stochastic Discount Factor (SDF), with the SDF characterising the representative agent's intertemporal marginal rate of substitution. Within this framework, cross-sectional return variation reflects compensation for systematic risk exposures, and equilibrium risk premia are determined by covariances of asset payoffs with the SDF. The Efficient Market Hypothesis (Fama, 1970) imposes the further restriction that asset prices fully reflect all available information, such that expected excess returns reflect only rational risk premia rather than predictable mispricings.

The present paper does not operate within the SDF-equilibrium paradigm. It does not derive the cross-section of expected returns from first principles, it does not identify the SDF, and it does not test whether observed return patterns are consistent with rational risk compensation. Instead, it operates within the empirical market-structure and systemic-risk monitoring literature, which Fama (1970) himself acknowledged as a legitimate complement to the equilibrium paradigm: the empirical

observation of return co-movements, tail dependencies, and regime changes may reveal information relevant to risk management without requiring a complete structural equilibrium model.

The departure from equilibrium assumptions is motivated by the empirical properties of financial return data. As documented by Cont (2001), actual asset returns exhibit heavy tails, volatility clustering, non-linear dependence, and regime changes that are inconsistent with the multivariate Gaussian return distributions assumed in classical portfolio theory. Campbell, Lo and MacKinlay (1997) and Tsay (2010) provide comprehensive treatments of these empirical departures. The implication, relevant to the present framework, is that linear correlation matrices — the workhorse of traditional portfolio optimisation — systematically mischaracterise the actual joint distribution of returns under stress conditions, motivating the network-topological, tail-dependence, and dynamic-correlation approaches employed here.

This paper is therefore best understood as an applied empirical contribution to the systemic risk monitoring literature, positioned alongside Kritzman et al. (2011) on the Absorption Ratio, Billio et al. (2012) on Granger-causality connectedness, and Diebold and Yilmaz (2014) on variance-decomposition connectedness. None of these papers derives systemic risk measures from equilibrium theory; all are positioned as empirical diagnostic frameworks. The present paper follows this tradition while extending it through methodological triangulation: the same substantive empirical questions are addressed using methodologically independent procedures (network topology, parametric DCC-GARCH, multiple-testing-corrected Granger, regime-switching HMMs), and findings are reported with explicit attention to which conclusions are robust across methods and which are method-specific.

2.2. Network Topology in Financial Markets

The application of network theory to financial markets originates with Mantegna (1999), who proposed the correlation-distance transformation $d(i,j) = \sqrt{2(1-\rho_{ij})}$ and the minimum spanning tree as a tool for revealing hierarchical structure in financial return data. Bonanno et al. (2003) showed that MST topology exhibits scale-free properties. Onnela et al. (2003) examined the temporal evolution of MST topology across crisis and non-crisis periods. Critically, these papers, and the present paper, interpret MST topology as a descriptive characterisation of the empirical correlation structure — not as evidence of specific economic transmission mechanisms. Network centrality and peripherality in the MST reflect the conditional correlation structure of returns; they do not, by themselves, establish economic causation.

2.3. Information Flow and Granger Causality

Billio et al. (2012) applied pairwise Granger causality to financial institutions and demonstrated that the resulting connectedness measures have predictive power for systemic risk indicators. Diebold and Yilmaz (2014) formalised the connection between forecast-error variance decompositions and network connectedness. Bouri et al. (2021) applied

these methods to multi-asset global portfolios during the COVID-19 period. Granger causality tests, as Granger (1969) himself emphasised, establish predictive precedence — whether lagged values of one series improve the forecast of another — rather than structural economic causation. The present paper applies Granger causality as a descriptive tool for identifying directional predictability patterns, with explicit acknowledgment of its econometric limitations (Section 3.4) and with both uncorrected and Bonferroni-corrected significance thresholds reported throughout (Section 4.3).

2.4. Dynamic Conditional Correlation Modelling

Engle's (2002) Dynamic Conditional Correlation (DCC) model is the standard parametric framework for time-varying multivariate correlation analysis. Building on the univariate GARCH(1,1) specification (Bollerslev, 1986), DCC estimates conditional correlations jointly with conditional variances in a two-stage procedure: stage one fits univariate GARCH on each asset to capture asset-specific volatility clustering (Cont, 2001); stage two models the correlation matrix of standardised residuals as a smoothly evolving function of past innovations. The DCC framework provides a parametric benchmark against which non-parametric topological characterisations can be compared. The DCC-GARCH model is employed in this capacity in Section 4.4, estimating a scalar DCC specification on the long-history sample and comparing the resulting end-of-sample correlation matrix to the static Pearson matrix used elsewhere in the analysis.

2.5. Regime-Switching Models

The regime-switching properties of financial markets have been studied since Hamilton (1989, 1994), who introduced Markov-switching models to capture nonstationary dynamics. Ang and Bekaert (2002) demonstrated that international equity correlations are systematically higher in bear regimes, with direct implications for multi-asset diversification. Guidolin and Timmermann (2007) identified four distinct multi-asset regimes with characteristically different return distributions. Bulla and Bulla (2006) demonstrated the superiority of hidden Markov models over threshold-based regime models in financial applications. The present paper applies a three-state Gaussian HMM with state count selected by the Bayesian Information Criterion (Section 3.6), following standard practice in the HMM literature. State count is held fixed at $K = 3$ for both the in-sample analysis (Section 4.7) and the out-of-sample evaluation (Section 4.7.4) to enable direct comparison between the two.

2.6. Tail Risk and Systemic Contagion

Longin and Solnik (2001) established that international equity correlations increase in down markets, with lower tail dependence coefficients providing a more accurate measure of contagion risk than full-distribution correlation. Adrian and Brunnermeier (2016) introduced CoVaR to quantify systemic risk through conditional Value-at-Risk. Acharya et al. (2017) proposed Marginal Expected Shortfall. Baur and McDermott (2010) documented the regime-conditional hedging

properties of gold. These contributions establish the econometric precedent for the tail-risk quantification layer of the present framework.

2.7. The Absorption Ratio

Kritzman et al. (2011) formalised the Systemic Absorption Ratio as a rolling PCA-based measure of market fragility, demonstrating across multiple historical crisis periods — including the 2008 Global Financial Crisis, the 2000 dot-com collapse, and the 1997 Asian crisis — that elevated AR values reliably precede and persist through significant market dislocations. A central insight of Kritzman et al.'s analysis is that the AR is a regime-persistence indicator rather than a contemporaneous crash detector: AR values rise as cross-asset correlation structure consolidates around a single dominant factor, and remain elevated for extended periods after acute crash dates as the consolidated structure dissipates only slowly.

2.8. Contribution Relative to Existing Literature

The present paper contributes to the empirical systemic-risk monitoring literature by integrating four established methodological streams — network topology, dynamic causality analysis with multiple-testing correction, tail-risk quantification, and HMM regime classification with out-of-sample evaluation — into a unified framework applied concurrently with an active stress event, and by providing methodological triangulation between non-parametric topological analysis and the standard parametric DCC-GARCH benchmark. The novelty lies in the integration, the volume-weighted MST extension, the out-of-sample HMM evaluation calibrated on a 14-year multi-crisis training history, and the multi-crisis Absorption Ratio replication.

3. Materials and Methods

3.1. Data and Sample

The analytical core uses daily adjusted closing prices for nine globally representative assets retrieved via the yfinance Python library over April 2025 to 24 April 2026, yielding $N = 261$ trading-day observations. For the out-of-sample HMM evaluation (Section 4.7.4) and the DCC-GARCH baseline (Section 4.4), a long-history sample over March 2011 to 24 April 2026 ($N = 3,472$ observations, all nine assets present continuously) is used. For the multi-crisis Absorption Ratio replication (Section 4.7.1), four historical windows are sliced: the Global Financial Crisis (June 2007 to December 2009; eight assets), COVID-19 dislocation (December 2019 to December 2020), Federal Reserve tightening (January 2022 to June 2023), and the current-study window. Table 1 details the full asset universe.

An important sample limitation must be stated at the outset. The 261-observation current-study window covers a single 13-month episode, which is short for systemic-risk modelling. The conclusions derived from this window are correspondingly exploratory and descriptive. They are not claimed to establish stable financial regularities generalising across multiple economic cycles. The multi-crisis AR replication (Section 4.7.1)

and the long-history DCC-GARCH (Section 4.4) and out-of-sample HMM (Section 4.7.4) analyses are intended precisely to address this single-episode concern.

Table 1. Asset universe, ticker symbols, and data sources. Role designations are descriptive labels reflecting the asset's role in the analysis; they are not equilibrium constructs.

Asset	Ticker	Source	Role in Analysis
US Equity (S&P 500)	^GSPC	Yahoo Finance	Equity cluster
European Equity (Stoxx 50)	^STOXX50E	Yahoo Finance	Equity cluster
Japanese Equity (Nikkei 225)	^N225	Yahoo Finance	Equity cluster
Chinese Equity (CSI 300)	000300.SS	Yahoo Finance	Equity cluster
Emerging Market Equity	EEM	Yahoo Finance	Equity cluster
Gold	GLD	Yahoo Finance	Hedge
US Dollar Index	DX-Y.NYB	Yahoo Finance	Macro anchor
Brent Crude Oil	BZ=F	Yahoo Finance	Macro anchor
US 10-Year Treasury Yield	^TNX	Yahoo Finance	Macro anchor

Source: Author's own work, based on data retrieved from Yahoo Finance.

Log returns are computed as $r_{\{i,t\}} = \ln(P_{\{i,t\}}/P_{\{i,t-1\}})$, with forward-fill applied prior to differencing to address non-synchronous trading. Macroeconomic context data (CPI; Federal Funds Rate) are retrieved from the Federal Reserve Economic Data (FRED) database. All assets are tested for unit roots using the Augmented Dickey-Fuller test with lag selection by the Akaike Information Criterion; the unit-root null is rejected at the 1% level for all nine assets on the log-return series, confirming stationarity.

3.2. Layer 1: Topological Analysis

3.2.1. Distance Transformation

Following Mantegna (1999), the sample Pearson correlation matrix C (9×9) is transformed into a metric distance space:

$$d(i,j) = \sqrt{2(1 - \rho_{\{ij\}})} \quad (1)$$

This transformation satisfies the three metric axioms (non-negativity, symmetry, and the triangle inequality) and produces the dissimilarity matrix D serving as common input to MDS, MST, and hierarchical clustering. The function $d(i,j)$ is a function of the sample Pearson correlation — a linear, second-moment summary statistic — and does not capture non-linear dependencies, higher-order co-moments, or conditional dependence structure (Cont, 2001). The resulting topological

objects therefore characterise the linear correlation geometry of returns within the sample period.

3.2.2. Multidimensional Scaling

Classical MDS is applied to D (scikit-learn, `n_components = 3`, `random_state = 42`). Quality is measured by Kruskal's Stress Score:

$$Stress = \sqrt{\sum_{i < j} (d_{ij} - \hat{d}_{ij})^2 / \sum_{i < j} d_{ij}^2} \quad (2)$$

A Stress value below 0.10 is conventionally considered excellent; below 0.20 acceptable. The empirical value of 0.11 reported in Section 4.2 indicates that three dimensions preserve approximately 89% of the original distance information.

3.2.3. Minimum Spanning Tree

The MST is constructed using NetworkX `minimum_spanning_tree` (Kruskal's algorithm) applied to the weighted graph G with edge weights $w_{ij} = d(i, j)$. Node degree and betweenness centrality are computed as descriptive hub indicators. A volume-weighted MST is constructed by replacing ρ_{ij} with volume-weighted correlations ρ^V_{ij} (Section 3.3.3).

3.2.4. Hierarchical Clustering

Ward's linkage method is applied to the condensed distance matrix D (`scipy.cluster.hierarchy.linkage`, `method = 'ward'`), minimising within-cluster variance at each merge step. The cluster labels assigned in the results section are descriptive ex-post characterisations of the resulting groups; they are not equilibrium constructs.

3.3. Layer 2: Dynamic Causality Analysis

3.3.1. Granger Causality Testing

Pairwise Granger causality tests are conducted across all 72 directional asset pairs ($i \rightarrow j$, $i \neq j$) using `statsmodels.grangercausalitytests` with `max_lag = 3`. A pair is classified as Granger-causally linked if $\min(p\text{-values across lags 1-3})$ is below the chosen significance threshold.

3.3.2. Bonferroni Correction

To address the multiple-testing problem inherent in 72 simultaneous tests, results are reported under two significance thresholds: an uncorrected threshold of $\alpha = 0.05$ and a Bonferroni-corrected threshold of $\alpha^* = 0.05/72 \approx 0.000694$. The Bonferroni correction controls the family-wise error rate at 5% under the assumption of test independence; given that these 72 tests share overlapping data and are not independent, Bonferroni is conservative — that is, surviving relationships are very likely real, but non-surviving relationships are not necessarily spurious.

3.3.3. Joint F-Test for Asset Exogeneity

A separate test of asset exogeneity is implemented via a full nine-asset Vector Autoregression at lag 3. For a candidate asset X , two joint F-tests are conducted: (i) the null that all eight non- X assets are jointly Granger-non-causal for X ('system $\rightarrow X$ '); and (ii) the null that X is jointly Granger-non-causal for the eight non- X assets (' $X \rightarrow$ system'). Forecast-error

variance decompositions at horizon 10 are computed to characterise the indirect transmission channels.

3.3.4. Lead-Lag Cross-Correlation

For each ordered pair (i, j) , the lead-lag cross-correlation function is computed at lags $k \in \{-5, \dots, +5\}$ days using NumPy's correlate function on standardised return series.

3.3.5. Volume-Weighted Correlation

The volume-weighted correlation matrix C^V is constructed by weighting each daily return observation by its normalised volume proxy $V_{\{i,t\}} / \text{mean}(V_i)$:

$$\rho^V_{\{ij\}} = \text{cov}_w(r_i, r_j) / (\sigma^V_i * \sigma^V_j) \quad (3)$$

For assets without volume data (indices, yields, currency proxies), the standard Pearson correlation is retained. The volume-weighted MST is constructed from $d^V(i,j) = \sqrt{2(1-\rho^V_{\{ij\}})}$.

3.4. Econometric Limitations of the Causality Analysis

This subsection explicitly addresses the econometric limitations of the Granger causality analysis. These limitations are stated upfront, and they inform the cautious language used throughout the paper in describing directional patterns.

First, Granger causality tests predictive precedence, not structural economic causation (Granger, 1969; Wooldridge, 2010). A finding that lagged values of asset i improve forecasts of asset j does not establish that i economically causes j ; the relationship may be spurious, driven by omitted common factors, or non-structural.

Second, the 261-observation current-study sample imposes significant finite-sample limitations. Finite-sample Granger tests are known to exhibit size distortions in small samples (Hamilton, 1994). All reported Granger findings should be treated as exploratory indicators.

Third, the multiple-testing problem is addressed explicitly through Bonferroni correction. Conducting 72 simultaneous pairwise tests at the 5% significance level implies an expected number of spurious rejections of approximately 3.6 under the global null hypothesis.

Fourth, the assumption of linear, constant-parameter Granger relationships is potentially violated in a sample spanning a structural break in market behaviour. Financial returns exhibit heteroskedasticity and non-linear dependence (Cont, 2001; Tsay, 2010). Standard Granger tests are derived under homoskedasticity and may be unreliable under GARCH-type volatility dynamics.

Fifth, parameter instability across the pre-stress and stress sub-periods is a potential concern. Sub-period analysis is identified as a priority extension for future research.

3.5. Layer 3: Tail Risk Quantification

3.5.1. Conditional Value-at-Risk

For each asset i , Value-at-Risk at confidence level $\alpha = 0.95$ is the empirical $(1-\alpha)$ -quantile:

$$VaR_i(\alpha) = -Q_{\{1-\alpha\}}(r_i) \quad (4)$$

CVaR (Expected Shortfall) is the mean of returns below the VaR threshold:

$$CVaR_i(\alpha) = -E[r_i | r_i < -VaR_i(\alpha)] \quad (5)$$

3.5.2. Lower Tail Dependence

The lower tail dependence coefficient is estimated non-parametrically (Longin and Solnik, 2001):

$$\lambda^L_{ij} = P(F_i(r_i) < u | F_j(r_j) < u) \quad (6)$$

estimated at $u = 0.05$. At $u = 0.05$ and $N = 261$, approximately 13 joint tail observations inform each bivariate estimate. These estimates carry substantial sampling uncertainty.

3.5.3. Volatility Sensitivity Coefficients

Sensitivity to the MOVE and VIX indices is estimated via OLS regression:

$$r_{i,t} = \alpha_i + \gamma_i \cdot \text{DELTA}(\text{MOVE}_t) + \epsilon_{i,t} \quad (7)$$

Newey-West standard errors with bandwidth equal to the integer ceiling of $N^{4/9}$ are employed to address potential serial correlation and heteroskedasticity (Wooldridge, 2010).

3.6. Layer 4: Regime Classification

3.6.1. Systemic Absorption Ratio

The rolling Systemic Absorption Ratio is computed over a 60-day window using StandardScaler-normalised returns and scikit-learn PCA:

$$AR_t = \lambda_1^{(t)} / \sum_{m=1}^9 \lambda_m^{(t)} \quad (8)$$

A threshold of $AR = 0.50$ is applied, above which the dominant factor accounts for more than half of total portfolio return variance (Kritzman et al., 2011).

3.6.2. HMM Specification and Model Selection

A Gaussian Hidden Markov Model is estimated using `hmmlearn`'s `GaussianHMM` class (`covariance_type = 'full'`, `n_iter = 200`, `random_state = 42`) on the feature vector:

$$X_t = [PC1_t, AR_t, \text{DELTA}_{AR_t}] \quad (9)$$

State count K is fixed at $K = 3$ in both the in-sample analysis and the out-of-sample evaluation. BIC comparison across $K \in \{2, 3, 4, 5\}$ is reported in Section 4.7.4 for transparency.

3.6.3. Out-of-Sample HMM Evaluation

The HMM analysis is extended via genuine out-of-sample evaluation. The model is fitted on the long-history training sample (March 2011 to March 2025, $N = 3,124$ observations). The fitted parameters — emission means, covariances, transition probabilities, and initial state distribution — are then frozen and applied to the test window (April 2025 to April 2026) without further re-estimation.

3.6.4. Regime Entropy

Regime classification confidence is validated through normalised Shannon entropy computed from the posterior probability vector:

$$H_t = - \sum_{k=1}^K P(S_t=k | O_{1:t}) * \log P(S_t=k | O_{1:t}) / \log K \quad (10)$$

3.7. Software and Reproducibility

All analyses are implemented in Python 3.10+ with: numpy, pandas, scikit-learn, scipy, statsmodels, hmmlearn, arch, networkx, matplotlib, seaborn, yfinance, and pandas-datareader. Random seeds are set to 42 throughout. No proprietary data are used. The complete annotated notebook and data retrieval pipeline are publicly available at <https://github.com/youness-yach/geometry-of-risk>.

4. Results

Throughout this section, empirical characterisations are expressed in descriptive terms consistent with the monitoring framework established in Section 2. Language implying structural economic causation, equilibrium pricing relationships, or universal financial regularities is avoided.

4.1. Return Distributions

Figure 1 presents daily log-return histograms for the nine assets. The distributions exhibit excess kurtosis and fat tails consistent with the stylised facts documented by Cont (2001). Oil displays the most pronounced tail behaviour (sample standard deviation approximately 0.030; minimum daily return -0.164). The USD Index has the lowest return volatility ($\sigma \approx 0.004$).

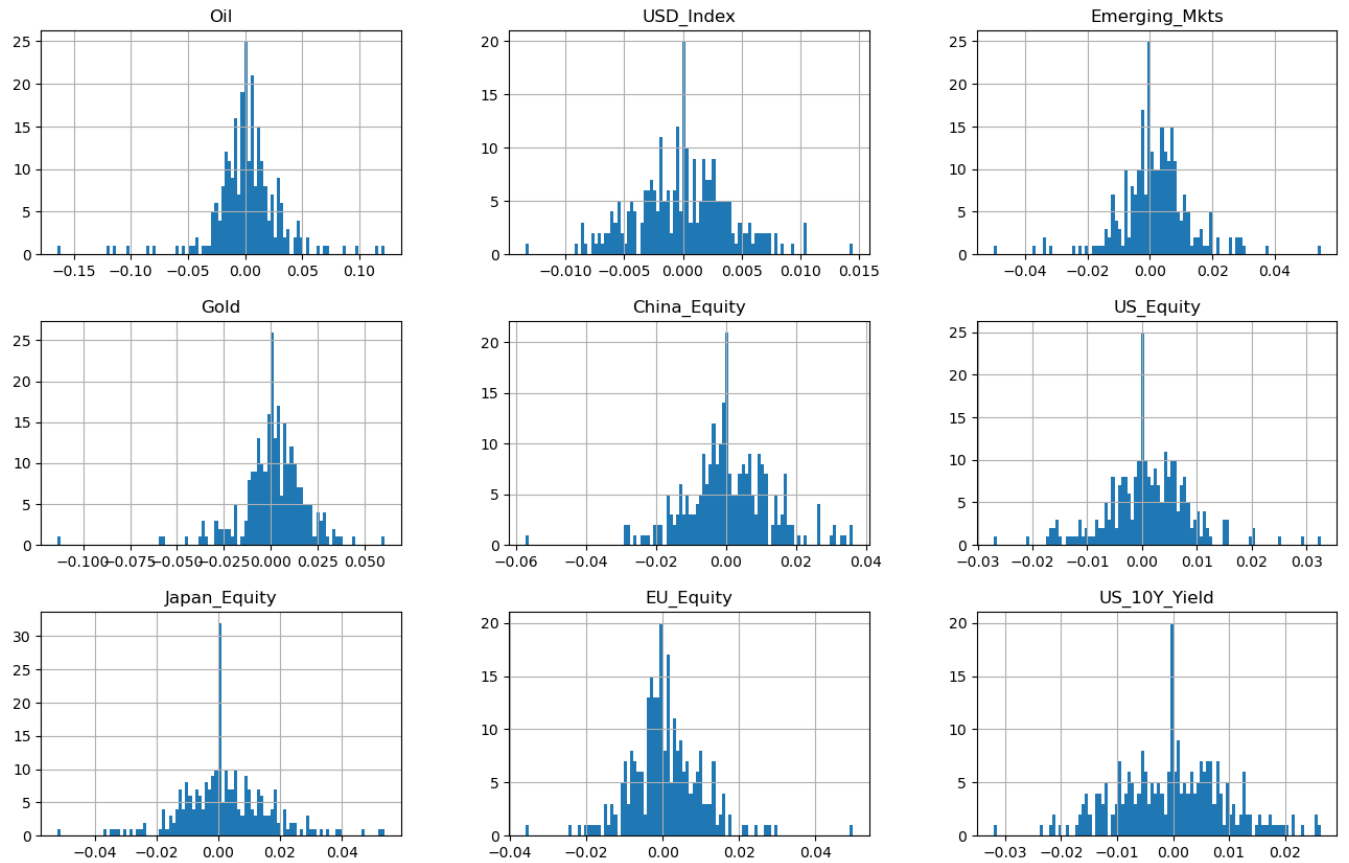


Figure 1. Daily log-return histograms, April 2025 to 24 April 2026 (N = 261). Non-normal, heavy-tailed distributions are observed across all assets, consistent with Cont (2001). *Source: Author's own work.*

4.2. Topological Analysis

4.2.1. Correlation Structure

Figure 2 presents the Pearson log-return correlation matrix. The highest pairwise correlations are observed for the China Equity–EM pair ($\rho = 0.72$) and the US Equity–EM pair ($\rho = 0.68$). The correlation matrix determinant is $\det(C) = 0.037$, indicating moderate multicollinearity.

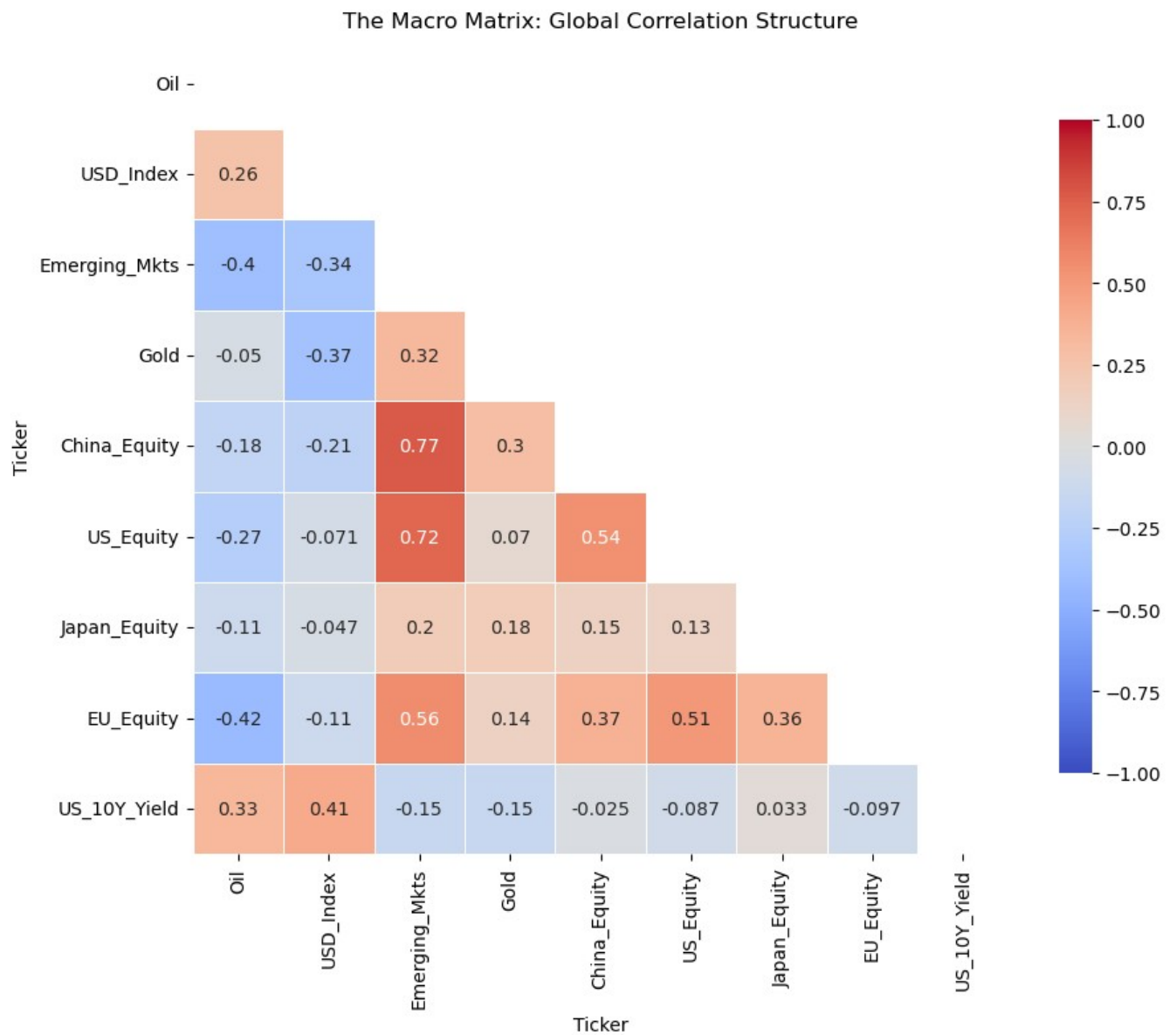


Figure 2. Pearson log-return correlation matrix, April 2025 to 24 April 2026.
Source: Author's own work.

4.2.2. Macroeconomic Context

Figure 3 presents a macroeconomic trend panel. The divergence between the US 10-Year Yield and the US Dollar Index from February 2026 reflects an empirically unusual configuration.

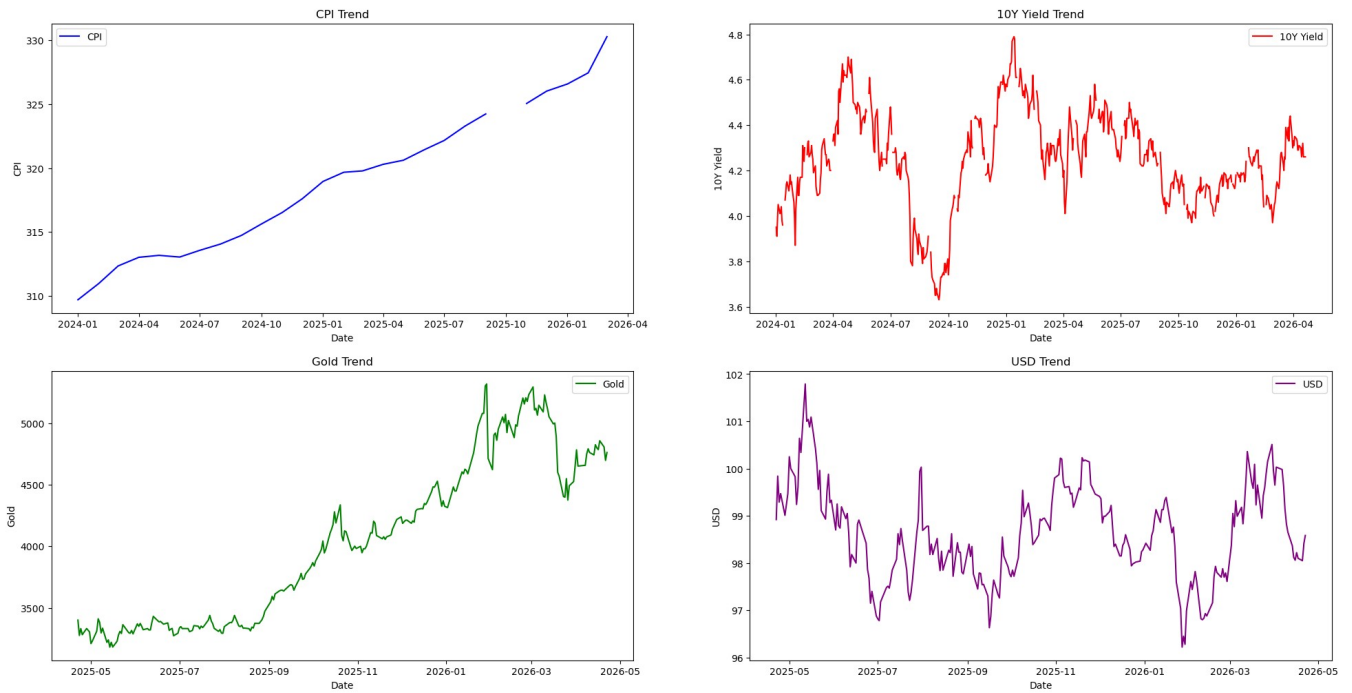
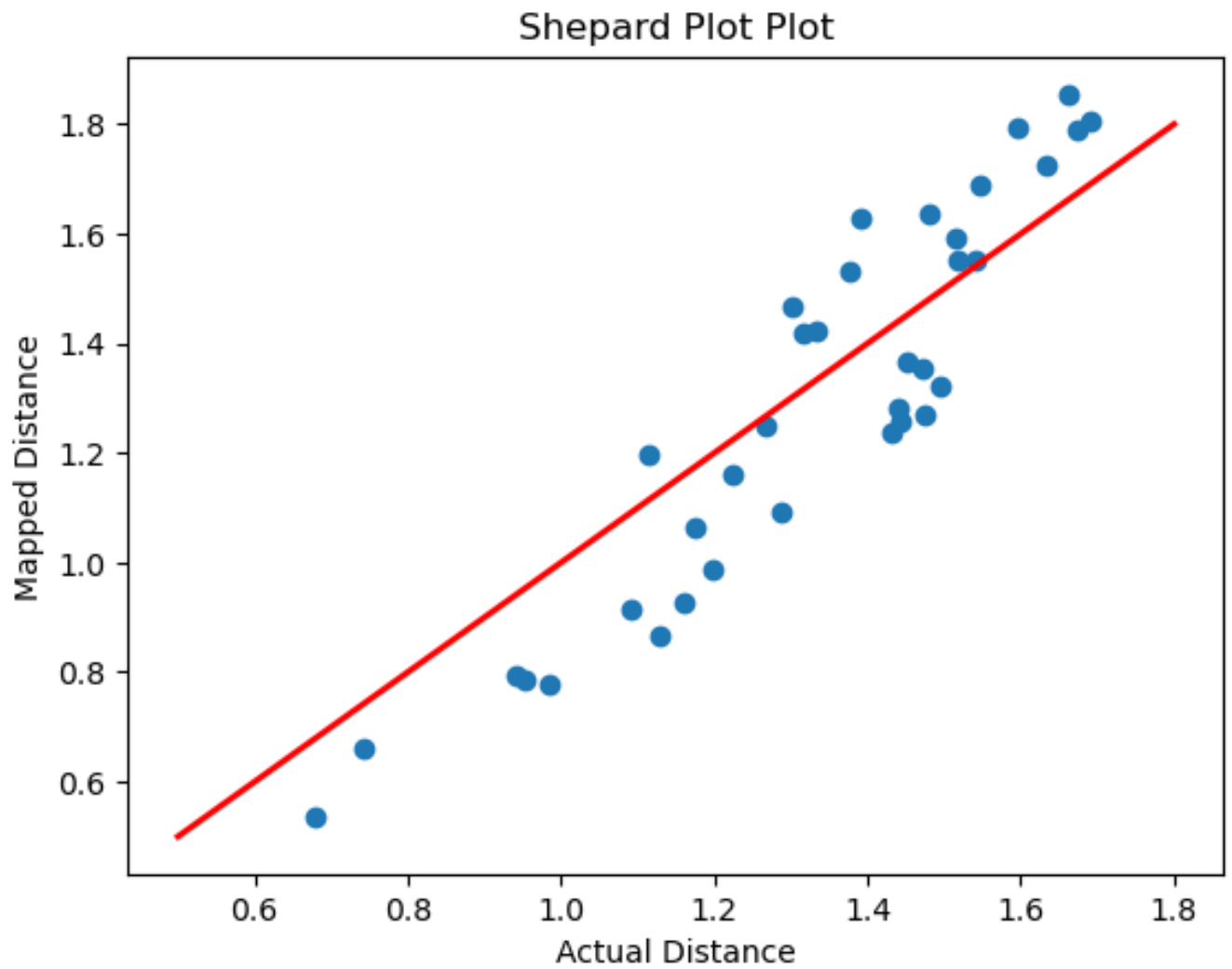


Figure 3. Macroeconomic context: US CPI, 10-Year Yield, USD Index, and Gold, April 2024 to April 2026. *Source: Author's own work.*

4.2.3. MDS and Hierarchical Clustering

Figure 4 shows the MDS distance preservation plot confirming a Kruskal Stress Score of 0.11 (approximately 89% information preservation). Figure 5 presents the Ward's method dendrogram. Three hierarchical groupings emerge: assets that merge at the lowest linkage distances (USD Index, US 10-Year Yield, Oil); a second grouping (US, EU, China, EM Equities); and a third grouping (Gold, Japan Equity). These groupings are labelled descriptively as Macro Anchors, Equity Cluster, and Hedge Cluster respectively.



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Figure 4. MDS distance preservation plot. Kruskal Stress Score = 0.11. *Source:* Author's own work.

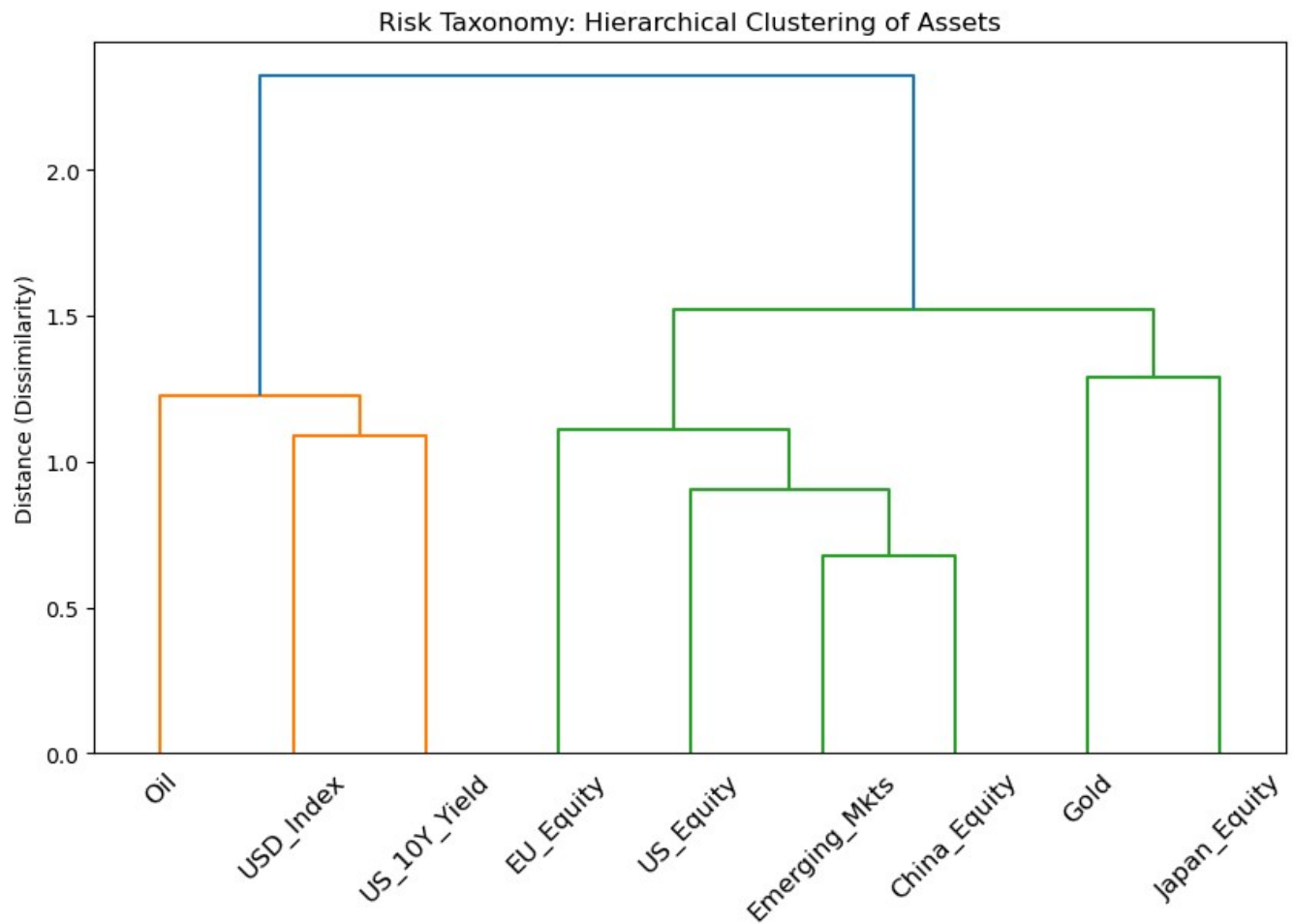


Figure 5. Ward's method hierarchical clustering dendrogram. *Source: Author's own work.*

4.2.4. Minimum Spanning Tree

Figure 6 presents the price-based Minimum Spanning Tree. Emerging Markets Equity exhibits the highest betweenness centrality (0.857), followed by US Equity (0.429). Japan Equity has the lowest betweenness centrality in the sample MST.

The Global Risk Skeleton (Minimum Spanning Tree)

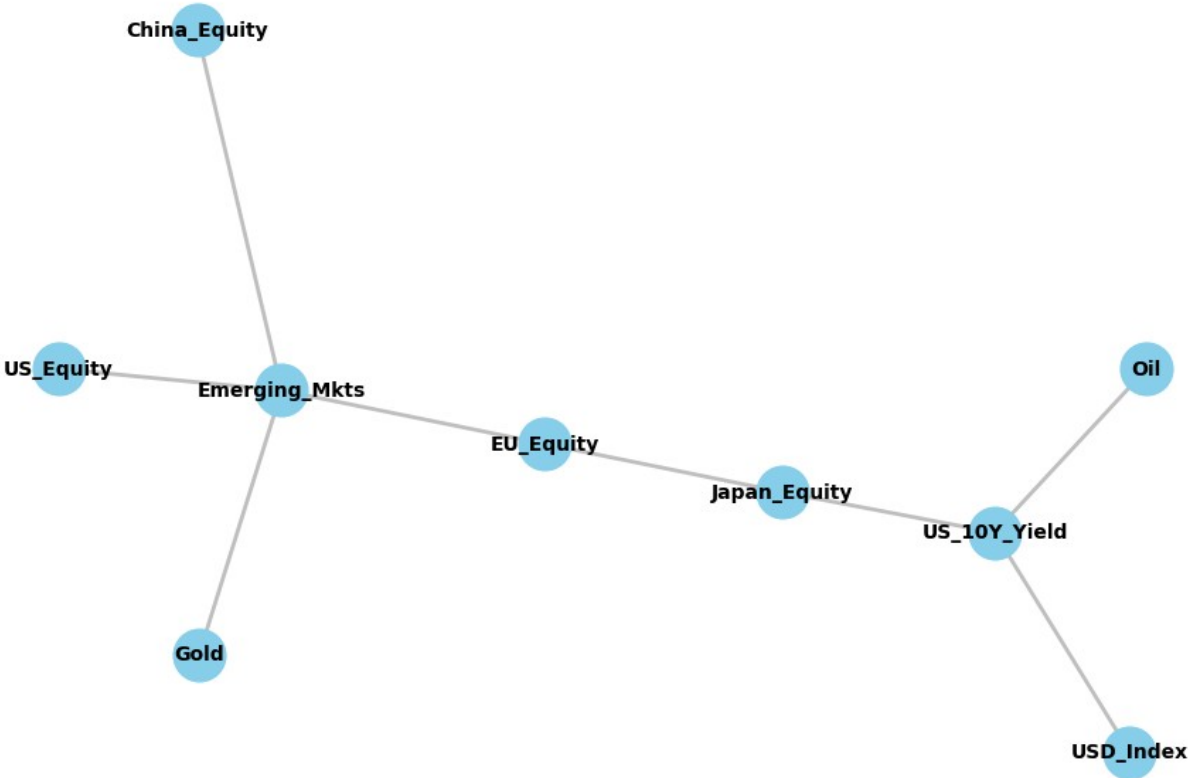


Figure 6. Price-based Minimum Spanning Tree, current-study window. Node diameter is proportional to betweenness centrality. *Source: Author's own work.*

4.3. *Dynamic Causality Analysis*

4.3.1. Volume-Weighted MST

Figure 7 presents the volume-weighted MST. Differences from the price-based topology include the repositioning of China Equity as a relatively central node connecting the macro-anchor cluster to the equity cluster, and the exclusive connection of Gold to the US 10-Year Yield.

The Global Risk Skeleton (Minimum Spanning Tree) Weighted

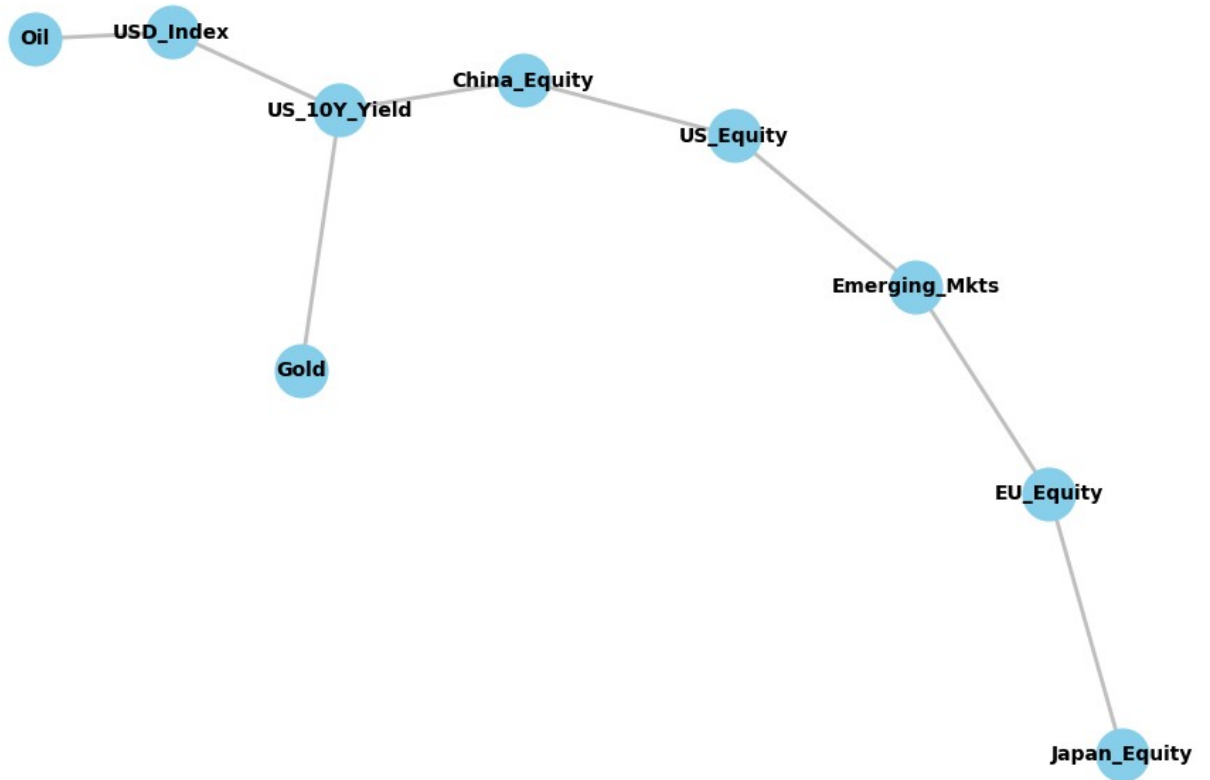


Figure 7. Volume-weighted MST, current-study window. *Source: Author's own work.*

4.3.2. Bonferroni-Corrected Granger Causality

Figure 8 presents the Granger causality significance matrix at two thresholds: uncorrected ($\alpha = 0.05$) and Bonferroni-corrected ($\alpha^* \approx 0.0007$). At $\alpha = 0.05$, the expected number of false positives is $72 \times 0.05 = 3.6$. After Bonferroni correction, only 4 of 72 directional relationships remain significant — and notably, all four point to the same asset:

Table 2. Bonferroni-corrected Granger causality survivors (current-study window, $N = 261$). All four surviving directional relationships point to Japan Equity.

Cause	Caused	p-value (min over lags 1–3)
Emerging Markets Equity	Japan Equity	$< 10^{-6}$
China Equity	Japan Equity	$< 10^{-6}$
US Equity	Japan Equity	$< 10^{-6}$
EU Equity	Japan Equity	$< 10^{-6}$

Source: Author's own work.

The Bonferroni-surviving evidence is therefore a single, narrow, and defensible claim: during the current-study window, the four major equity blocs (US, EU, China, Emerging Markets) collectively Granger-predict Japan Equity at the most stringent multiple-testing-corrected significance level.

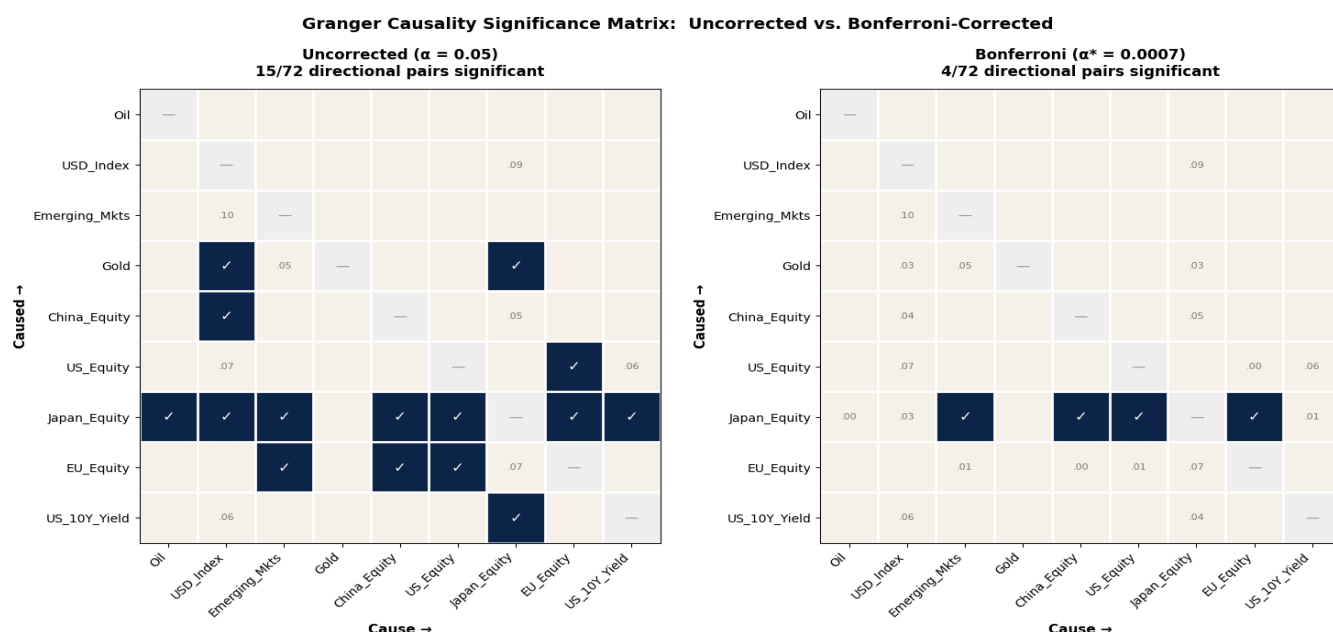


Figure 8. Granger causality significance matrix: uncorrected (left) versus Bonferroni-corrected (right). At the Bonferroni threshold, only 4 of 72 remain significant, and all four point to Japan Equity. *Source: Author's own work.*

4.3.3. Oil Exogeneity Tests

Pairwise Granger tests on Oil produce zero significant predictors and predictees at the Bonferroni-corrected threshold. To address Oil's exogeneity directly, two complementary joint F-tests in a full nine-asset VAR(3) specification are reported.

Table 3. Oil exogeneity: joint F-tests in a 9-asset VAR(3) specification. Both tests fail to reject the null.

Test	F-statistic	p-value	Decision
System $\rightarrow$ Oil (do the other 8 assets jointly Granger-cause Oil?)	0.78	0.765	Fail to reject null
Oil $\rightarrow$ System (does Oil jointly Granger-cause the other 8 assets?)	1.18	0.252	Fail to reject null

Source: Author's own work.

Despite the negative Granger results, the variance decomposition reveals an asymmetric coupling. At the 10-day horizon, Oil's own forecast-error variance is approximately 92.7% self-explained. Yet Oil contributes 20.9% to the 10-day forecast-error variance of EU Equity, 19.6% to Emerging Markets, and 15.8% to the US 10-Year Yield. Oil therefore operates as a quasi-exogenous asset whose own returns are

poorly explained by the financial cross-section, but whose shocks transmit to other assets through volatility-channel effects (Hamilton, 1994).

4.3.4. Relative Rotation Graphs

Figure 9 presents the Relative Rotation Graph relative to the S&P 500 benchmark. Japan Equity and Emerging Markets occupy the Leading quadrant, while the USD Index and Oil occupy the Lagging quadrant.

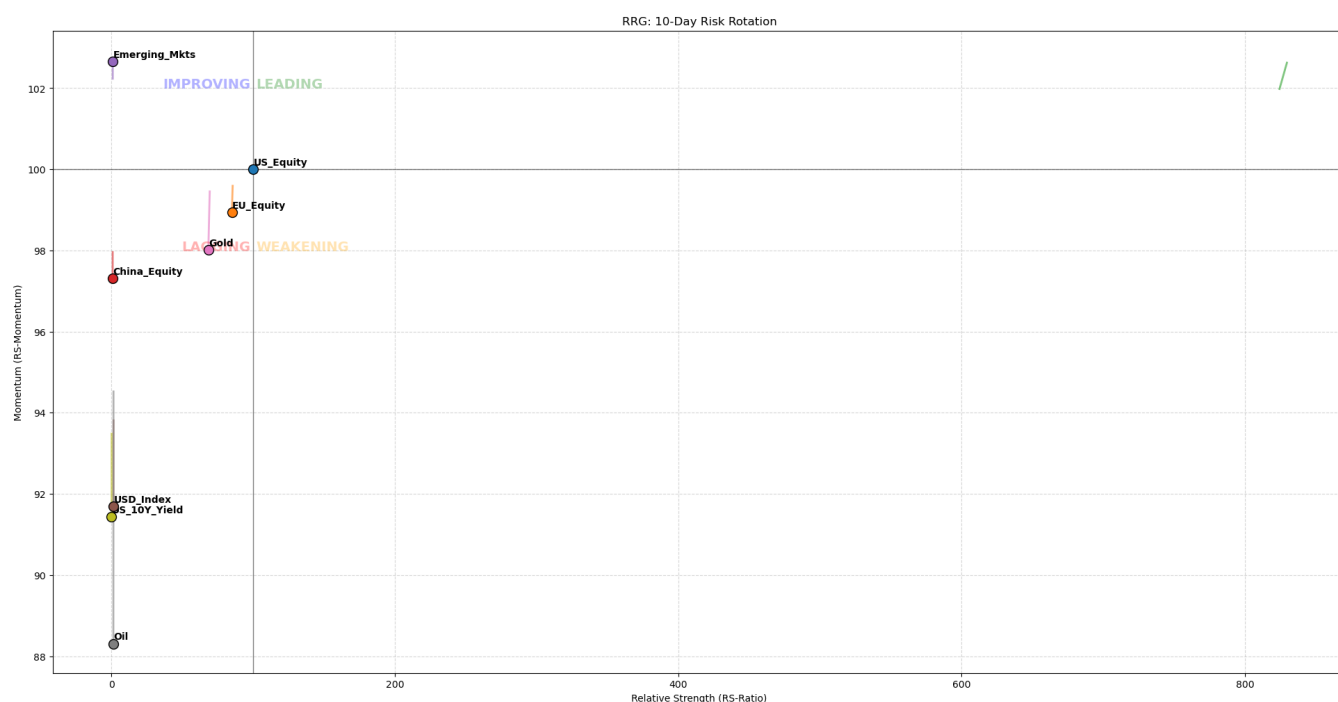


Figure 9. Relative Rotation Graph (RRG) relative to S&P 500. *Source: Author's own work.*

4.4. Dynamic Conditional Correlation Analysis

To provide a parametric benchmark for the network-topological analysis, a scalar Dynamic Conditional Correlation GARCH model (Engle, 2002) is estimated on the long-history sample (March 2011 to April 2026, $N = 3,472$).

4.4.1. Univariate GARCH(1,1) Estimation

Stage one fits a GARCH(1,1) specification to each of the nine assets. All nine fits converge with persistence parameters ($\alpha + \beta$) ranging from 0.953 to 0.999, consistent with Cont (2001).

Table 4. GARCH(1,1) parameter estimates, long-history sample (March 2011–April 2026, $N = 3,472$). Returns expressed in percent.

Asset	ω	α	β	$\alpha + \beta$
Oil	0.140	0.102	0.879	0.981
USD Index	0.001	0.036	0.958	0.994
EM Equity	0.083	0.124	0.832	0.956

Gold	0.021	0.052	0.933	0.984
China Equity	0.098	0.091	0.878	0.969
US Equity	0.046	0.166	0.797	0.963
Japan Equity	0.091	0.127	0.827	0.953
EU Equity	0.053	0.113	0.855	0.969
US 10Y Yield	0.016	0.053	0.946	0.999

Source: Author's own work.

4.4.2. DCC Parameter Estimation

Stage two estimates the scalar DCC parameters (a , b) by Quasi Maximum Likelihood on the standardised residuals from stage one. The fitted parameters are $a = 0.0136$ (innovation weight) and $b = 0.9749$ (persistence weight), giving total persistence $a + b = 0.989$. The implied half-life of cross-asset correlation shocks is approximately 27 trading days.

4.4.3. Time-Varying Correlations

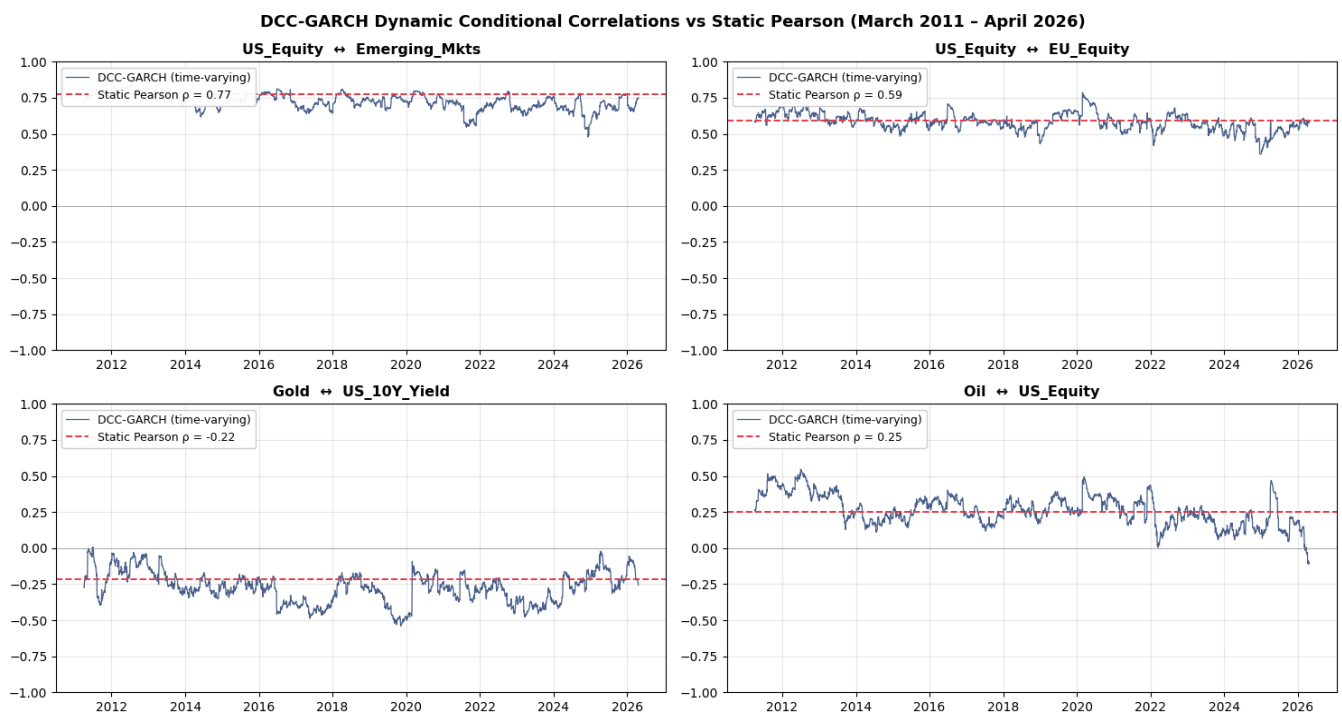


Figure 10. DCC-GARCH dynamic conditional correlations versus static Pearson correlation, March 2011 to April 2026. Four representative pairs are shown.

Source: Author's own work.

4.4.4. MST Comparison

Comparison of the static Pearson MST against the end-of-sample DCC-GARCH MST yields a Jaccard similarity of 0.455 across 8 MST edges. Five edges are shared between the two methods, forming the robust skeleton of the cross-asset network: China–EM, EM–EU, EM–US, USD–US 10Y Yield, and EU–Japan. The convergence of two methodologically

independent procedures on Japan Equity's centrality constitutes a more robust empirical signal than either method provides individually.

Table 5. Betweenness centrality comparison between static Pearson MST and DCC-GARCH end-of-sample MST. DCC promotes Japan Equity, China Equity, EU Equity, and the US 10-Year Yield to higher centrality.

Asset	Pearson MST Betweenness	DCC MST Betweenness
Emerging Markets Equity	0.857	0.607
US Equity	0.429	0.000
EU Equity	0.250	0.571
US 10-Year Yield	0.250	0.464
Japan Equity	0.000	0.536
China Equity	0.000	0.250
Oil	0.000	0.000
Gold	0.000	0.000
USD Index	0.000	0.000

Source: Author's own work.

4.5. Spectral Decomposition

Figure 10 presents the PCA scree plot. PC1 accounts for approximately 36% of total portfolio variance, below the 50% threshold associated with single-factor macro dominance (Kritzman et al., 2011). The combined explanatory power of PC1 and PC2 (36% + 18% = 54%) crosses this threshold.

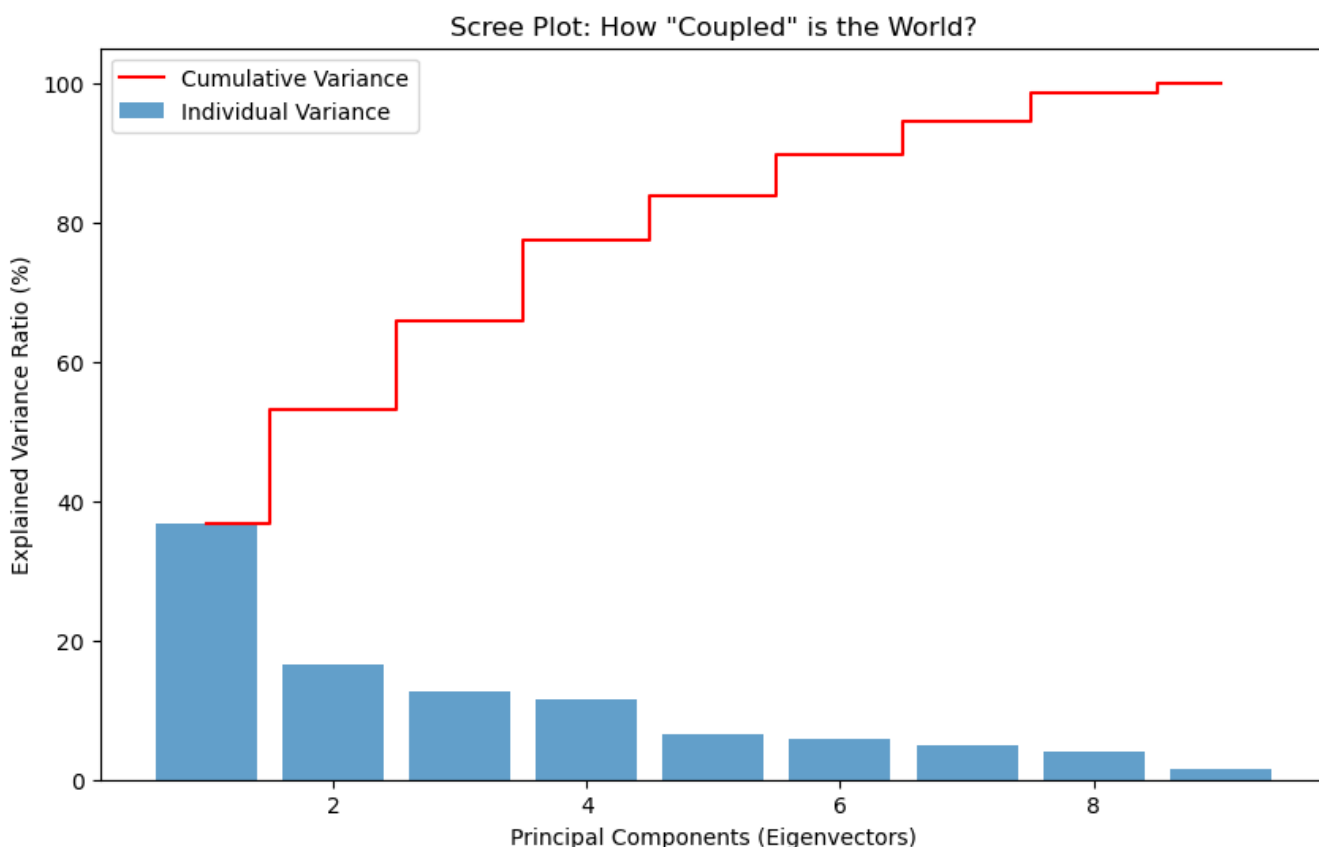


Figure 11. PCA scree plot. *Source: Author's own work.*

4.6. Tail Risk Quantification

4.6.1. CVaR and Tail Ratios

Table 6 presents Conditional Value-at-Risk and tail ratio estimates. Oil exhibits the highest tail ratio ($CVaR/VaR = 2.12$).

Table 6. CVaR and tail ratio estimates (95% confidence level), non-parametric estimation.

Asset	VaR(95%)	CVaR(95%)	Tail Ratio	Interpretation
Oil	-0.0345	-0.0734	2.12	Most extreme tail ratio
EM Equity	-0.0138	-0.0248	1.80	Second-highest equity tail ratio
Japan Equity	-0.0177	-0.0286	1.62	Wider tail than developed peers
Gold	-0.0275	-0.0427	1.55	Moderate
China Equity	-0.0185	-0.0259	1.40	Moderate
EU Equity	-0.0133	-0.0185	1.39	Moderate
US Equity	-0.0123	-0.0165	1.35	Lower equity tail ratio
US 10Y Yield	-0.0154	-0.0196	1.27	Near-Gaussian behaviour
USD Index	-0.0065	-0.0081	1.25	Most Gaussian-like

Source: Author's own work.

4.6.2. Tail Dependence

Figure 11 presents the lower tail dependence coefficient matrix. The China–EM–EU equity sub-group exhibits the highest estimated pairwise tail dependence coefficients ($\lambda^L > 0.35$).

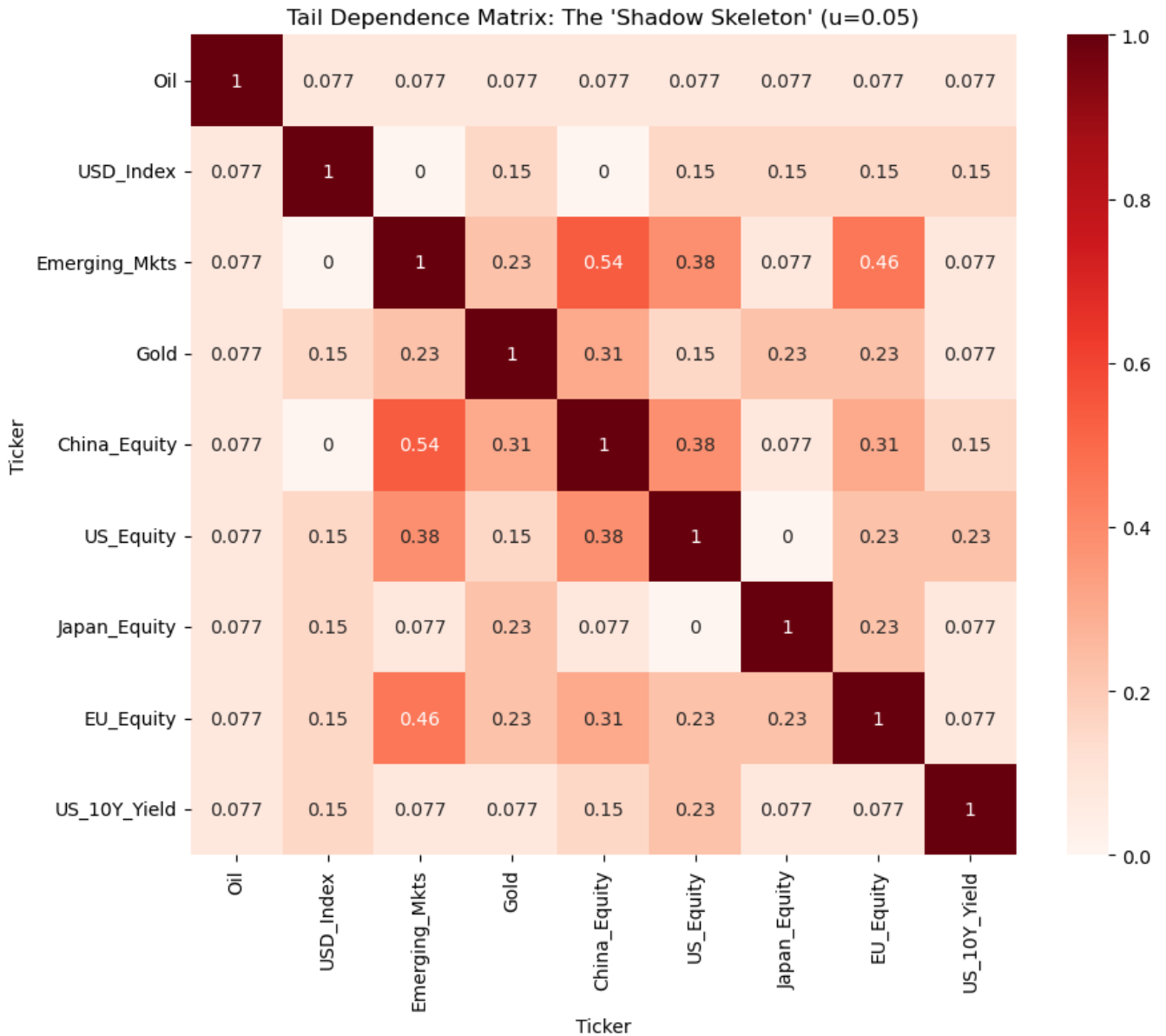


Figure 12. Lower tail dependence coefficient matrix ($u = 0.05$). Source: Author's own work.

4.6.3. Volatility Sensitivity

Figures 12, 13, and 14 present the MOVE and VIX sensitivity coefficients and the VIX/MOVE ratio. Oil exhibits the largest positive MOVE coefficient. US Equity and EM Equity exhibit the most negative MOVE coefficients.

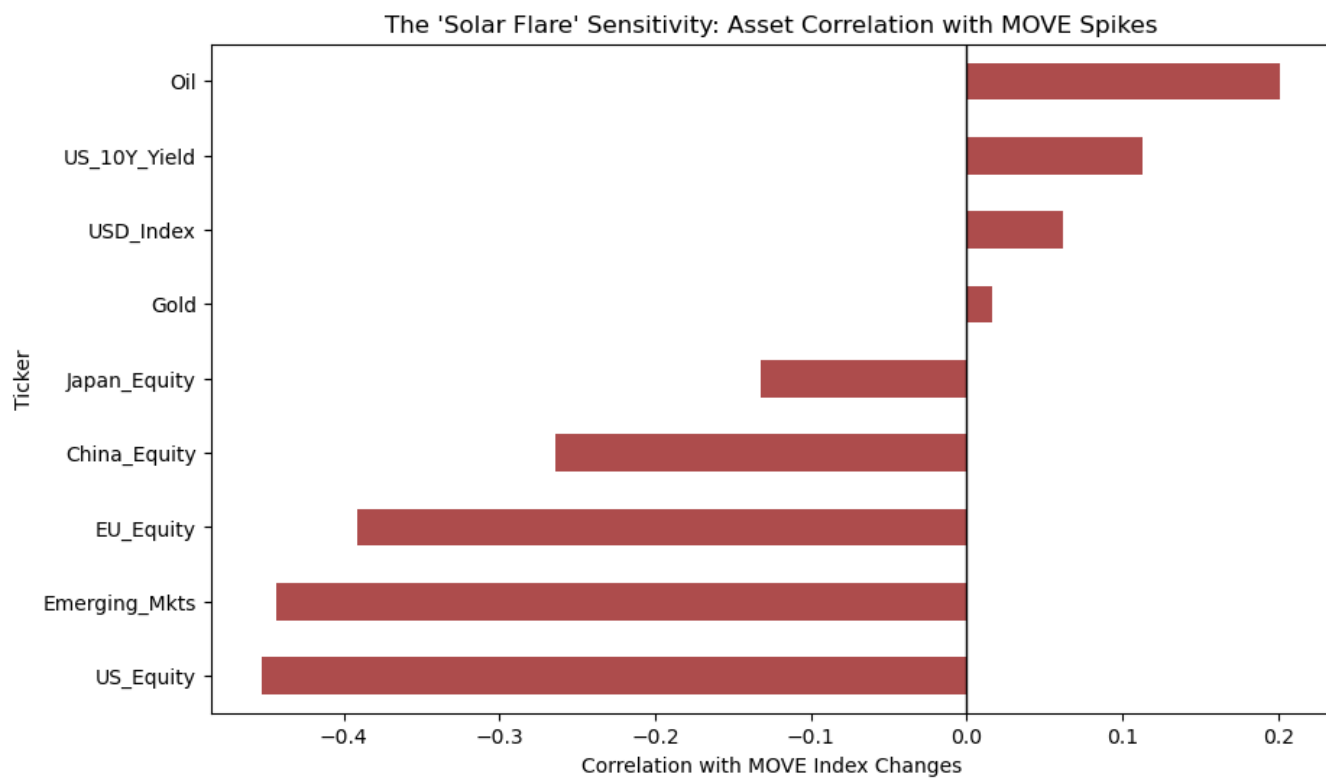


Figure 13. MOVE Index sensitivity coefficients by asset. *Source: Author's own work.*

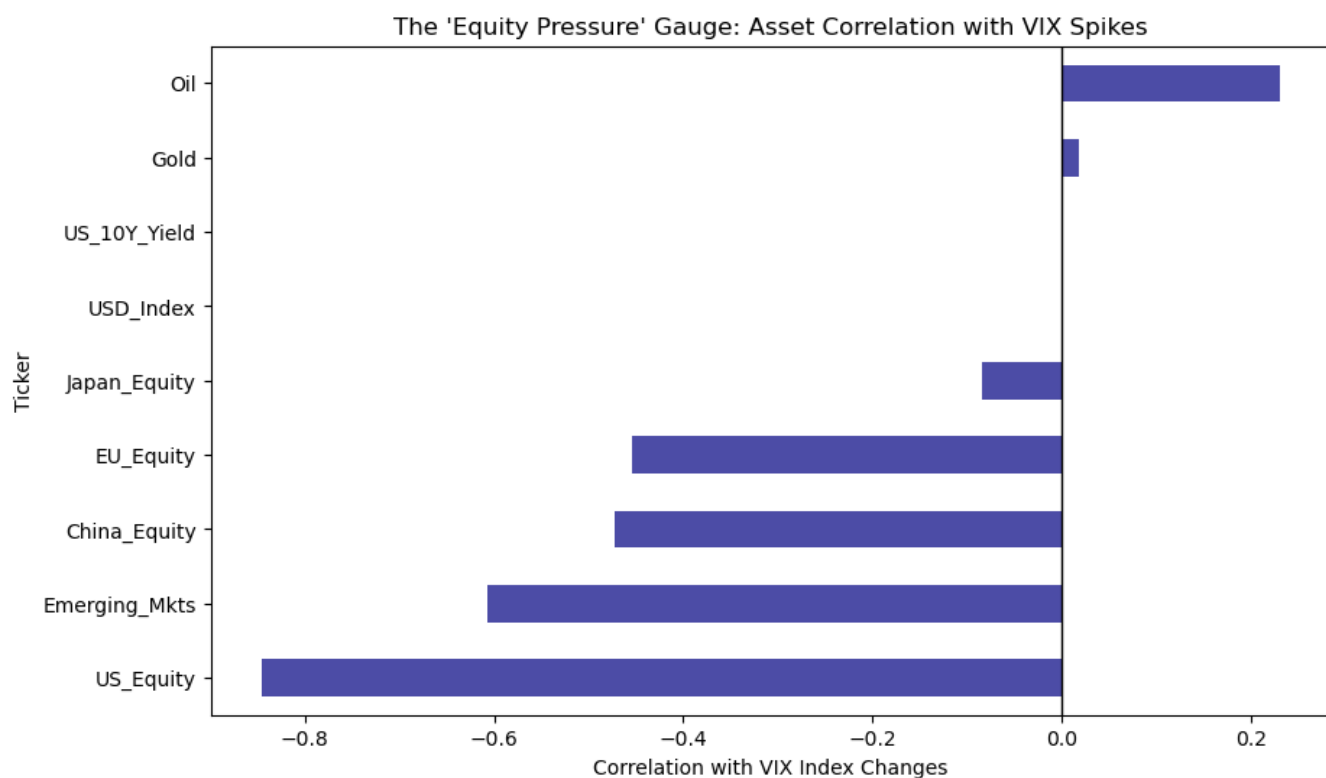


Figure 14. VIX sensitivity coefficients by asset. *Source: Author's own work.*

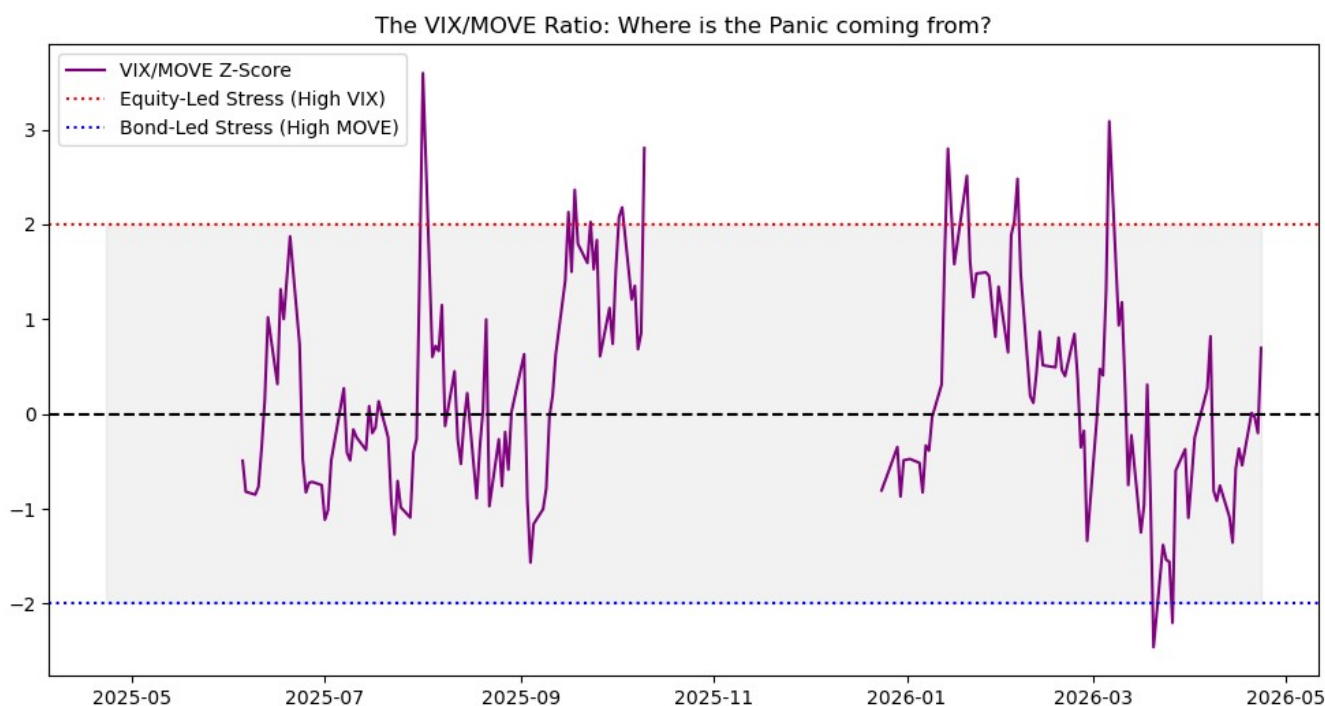


Figure 15. VIX/MOVE ratio time series. *Source: Author's own work.*

4.7. Regime Classification

4.7.1. Multi-Crisis Absorption Ratio Replication

To address concerns about single-episode evidence, the rolling 60-day Absorption Ratio is computed across four crisis windows. Figure 17 presents the four AR time series, and Table 7 summarises the per-window statistics.

Table 7. Multi-crisis Absorption Ratio replication. The 0.50 threshold is breached in all four episodes.

Crisis Window	N Assets	N Days	Mean AR	Max AR	% Days $\geq$ 0.50
GFC (2007–2009)	8	612	0.430	0.537	10.5%
COVID (2019–2020)	9	220	0.441	0.601	5.5%
Fed Tightening (2022–2023)	9	327	0.411	0.529	8.3%
Current Study (2025–2026)	9	217	0.357	0.507	1.4%

Source: Author's own work.

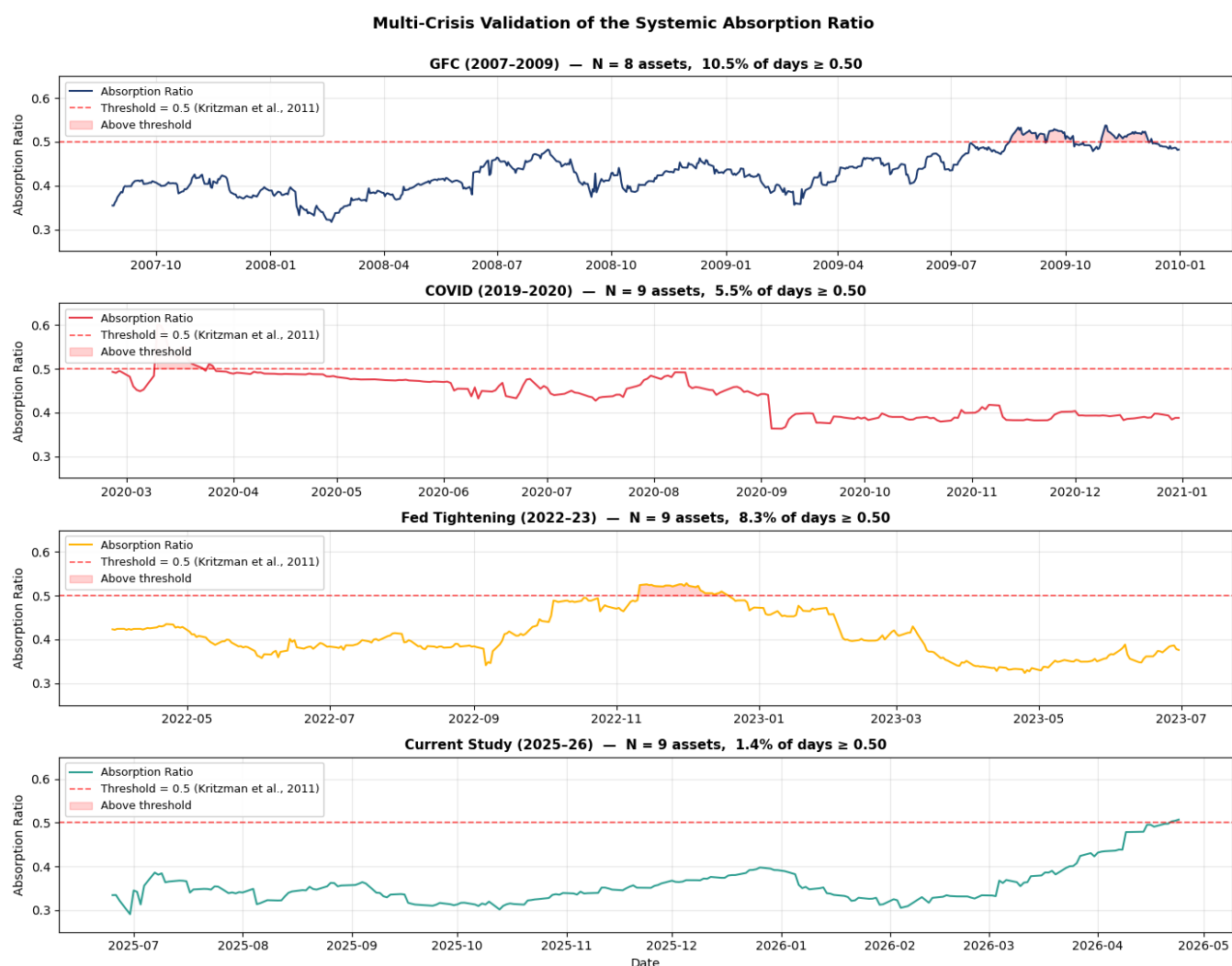


Figure 16. Multi-crisis validation of the rolling 60-day Systemic Absorption Ratio. The 0.50 threshold of Kritzman et al. (2011) is breached in all four episodes. *Source: Author's own work.*

Two methodological observations are required. First, the AR's temporal alignment with acute crash dates is loose: the AR peaks in the GFC window occur in late 2009, well after the September 2008 Lehman shock. This is consistent with the original findings of Kritzman et al. (2011): the AR is a regime-persistence indicator rather than a contemporaneous crash detector. Second, the current-study window exhibits the lowest mean AR of the four crises tested.

4.7.2. Rolling Absorption Ratio (Current Study)

Figure 16 presents the rolling 60-day Absorption Ratio for the current-study window. The AR ranges from approximately 0.29 (early 2026) to 0.51 (mid-April 2026), with the rise through the 0.50 threshold occurring in the last weeks of the sample.

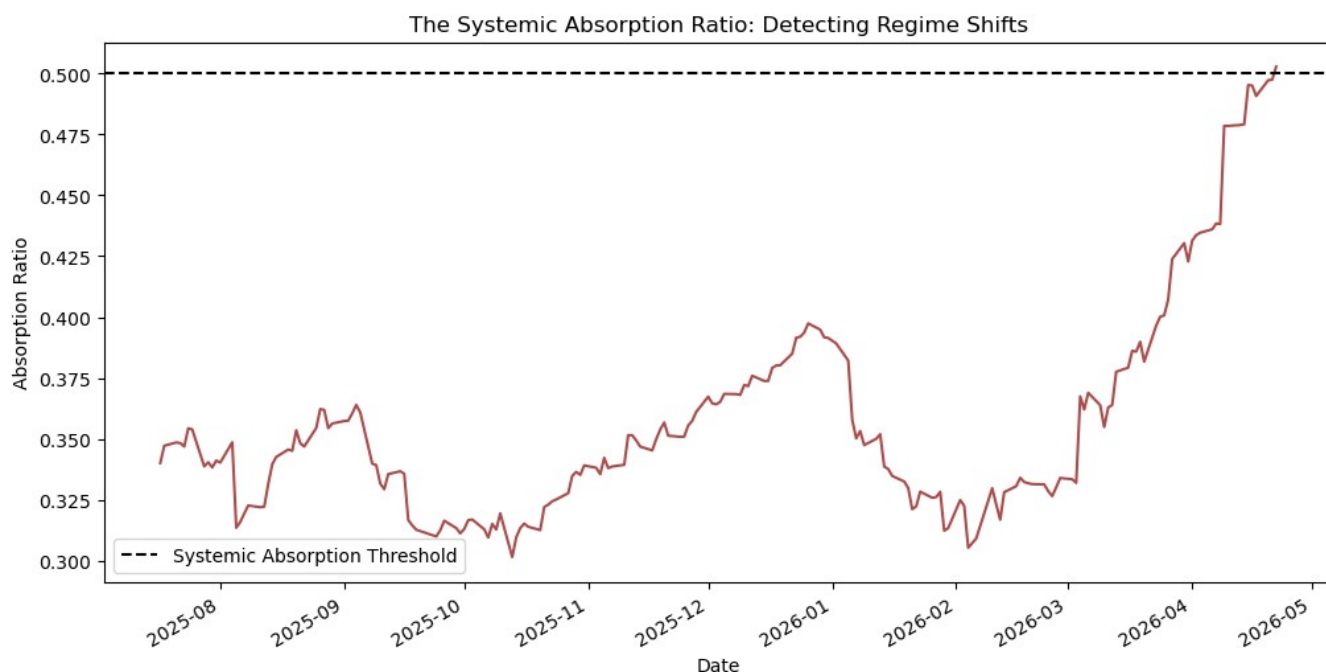


Figure 17. Rolling 60-day Systemic Absorption Ratio, current-study window. *Source: Author's own work.*

4.7.3. Standardised AR Delta

Figure 18 presents the standardised 15-day AR delta. The large negative delta observed in late April 2026 (-1.36 standard deviations) indicates a deceleration in the rate of AR increase.

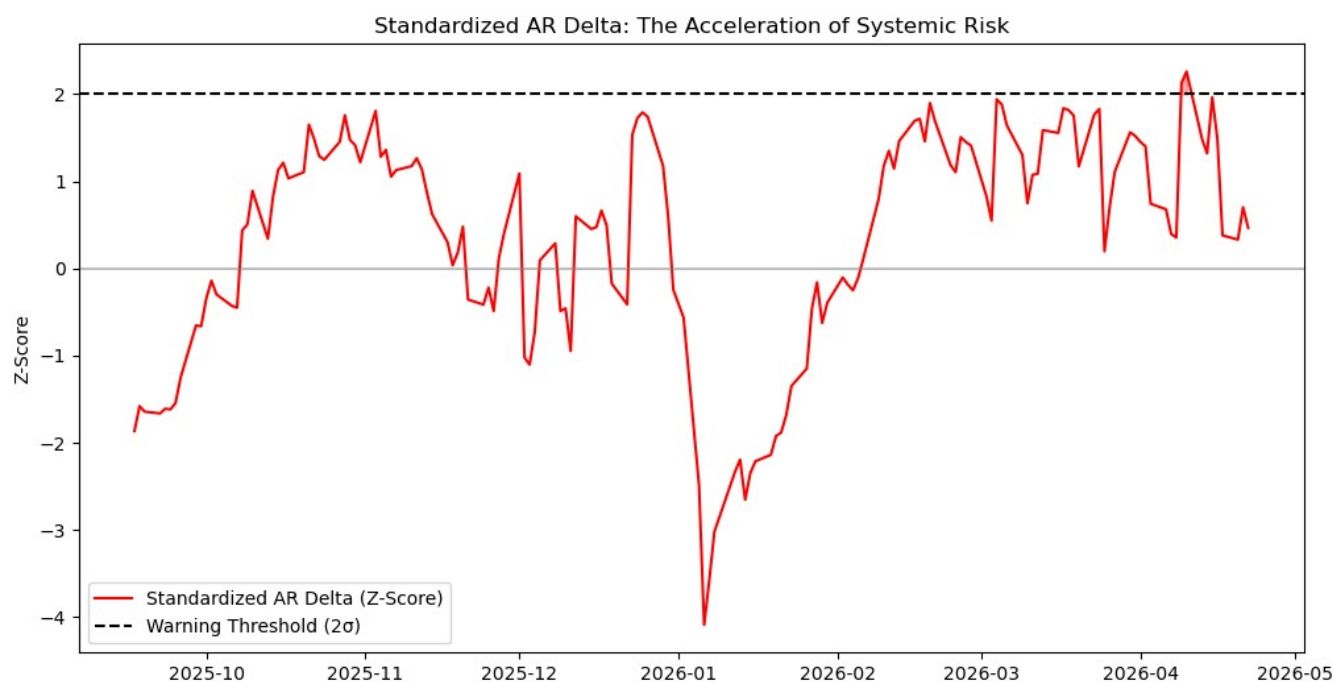


Figure 18. Standardised 15-day AR Delta, current-study window. *Source: Author's own work.*

4.7.4. Hidden Markov Model: In-Sample and Out-of-Sample

Figure 19 presents the first principal component returns. Figure 20 presents the in-sample HMM three-state regime classification. Table 8 summarises the BIC model selection over $K \in \{2, 3, 4, 5\}$.

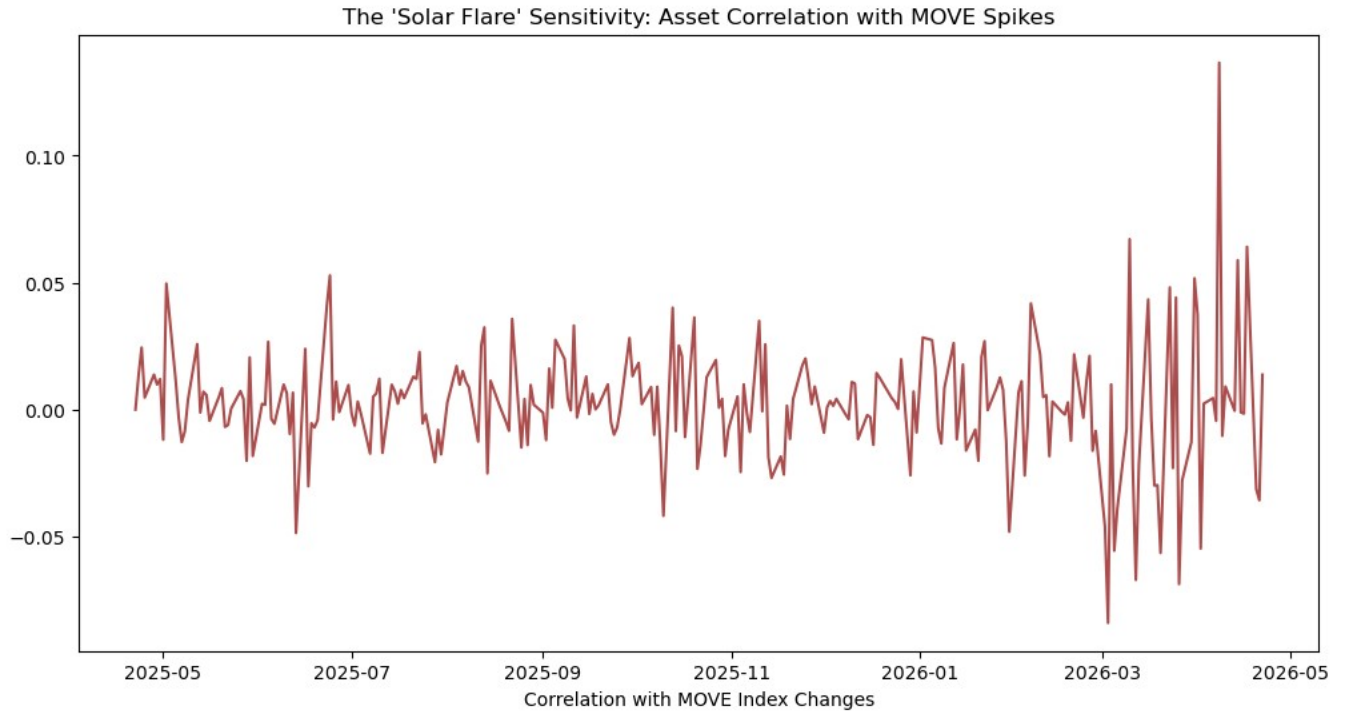


Figure 19. First principal component (PC1) return series. *Source: Author's own work.*

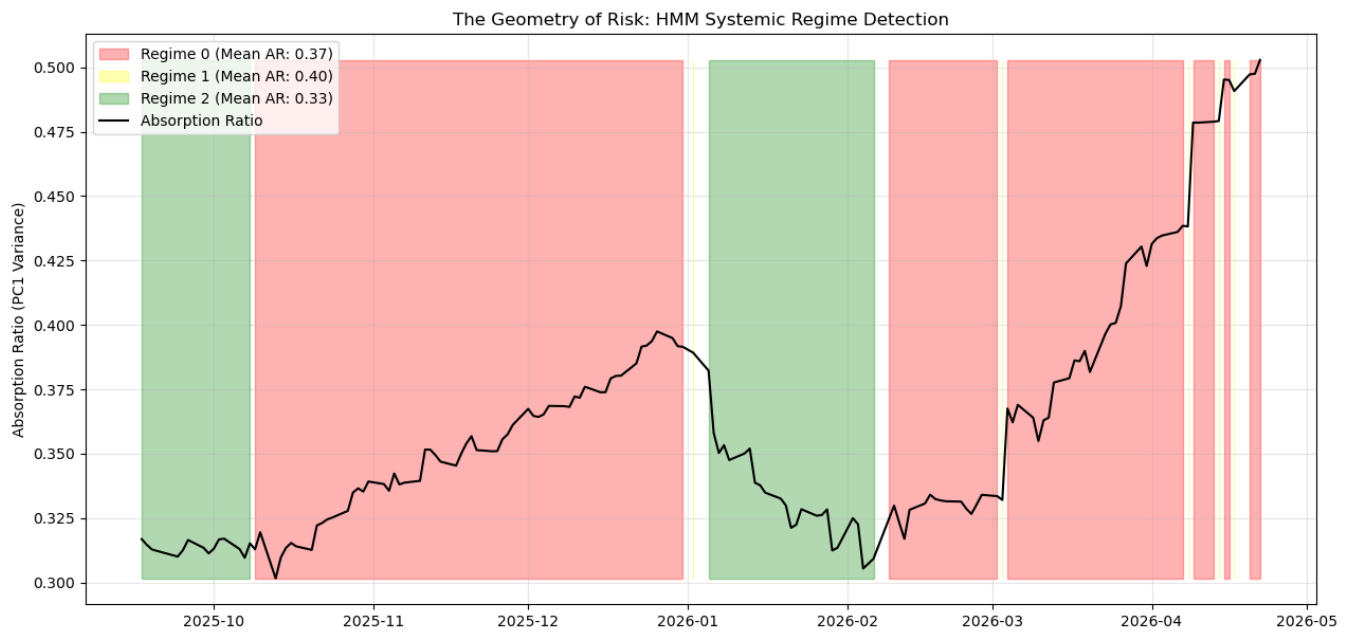


Figure 20. In-sample HMM three-state regime classification. *Source: Author's own work.*

Table 8. HMM BIC model selection on the long-history training sample (March 2011–March 2025, $N = 3,124$). $K = 3$ is retained for interpretive tractability.

States (K)	Log-Likelihood	Parameters (k)	BIC
2	7,340	21	−14,511
3	8,602	35	−16,923
4	8,447	51	−16,484
5	9,935	69	−19,314

Source: Author's own work.

Figure 21 presents the out-of-sample HMM regime classification. The 14-year-trained HMM identifies three states with separated AR means: Stable (mean AR ≈ 0.349), Transitional (mean AR ≈ 0.412), and Stress (mean AR ≈ 0.523). Applied to the test window without re-estimation, the framework activates the stress state on 10 of 244 test-period days (4.1%), including 5 consecutive days at the end of the test window (20–24 April 2026). Mean out-of-sample posterior entropy is 0.074 (range $[0, 0.63]$).

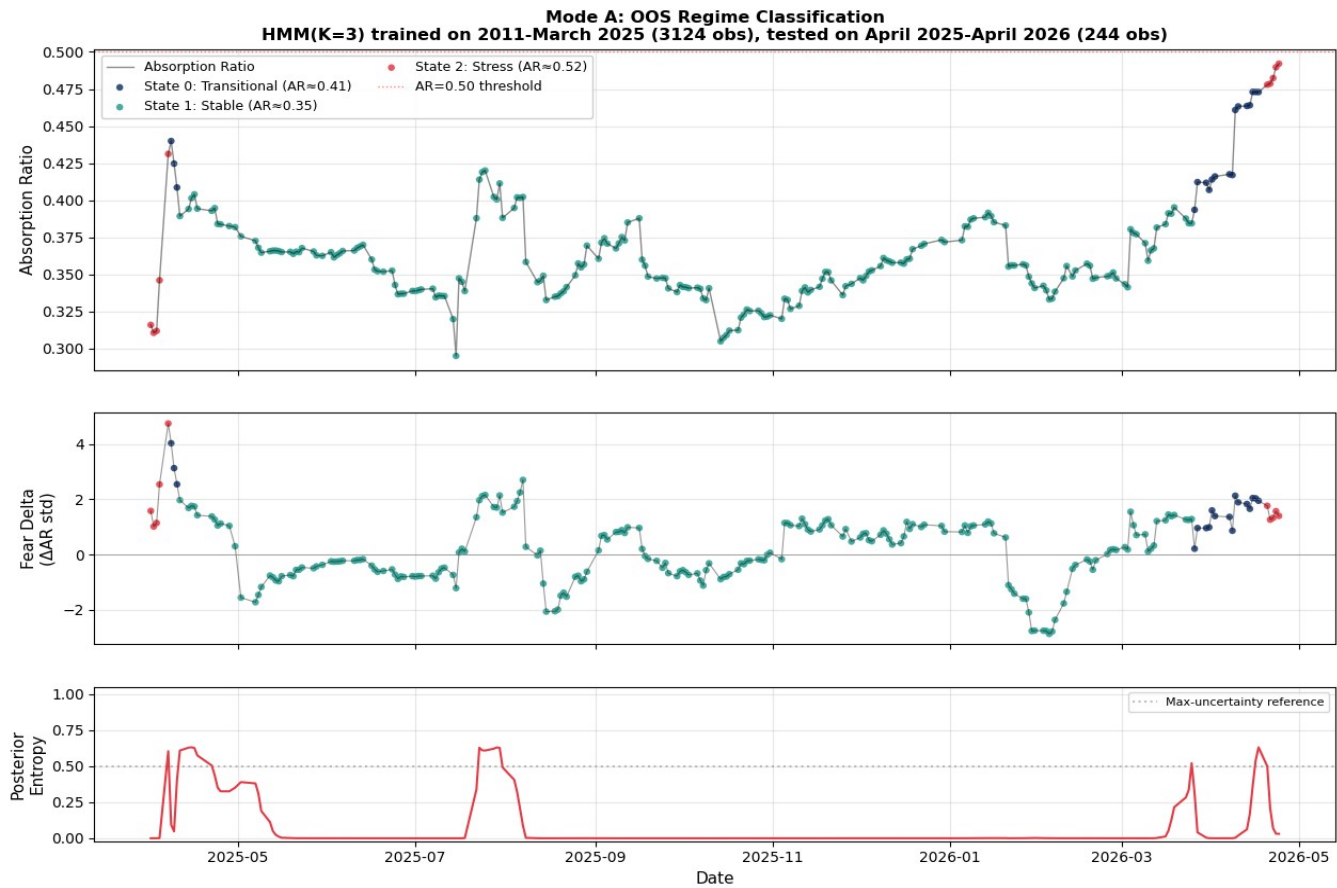


Figure 21. Out-of-sample HMM regime classification. The model is trained on the 14-year long-history sample (March 2011–March 2025) and applied to the test window without further re-estimation. Source: Author's own work.

A complementary base-rate diagnostic compares the training-period and test-period stress regime frequencies. In the 14-year training sample, the stress state accounts for approximately 25.6% of observations. The test-period stress frequency of 4.1% is therefore considerably lower than the training base rate, consistent with characterising April 2025–April 2026 as a relatively calm year by historical standards.

4.7.5. Regime Entropy

Figure 22 presents the regime entropy time series. Posterior entropy declines from elevated values during regime transitions to near-zero values during regime persistence.

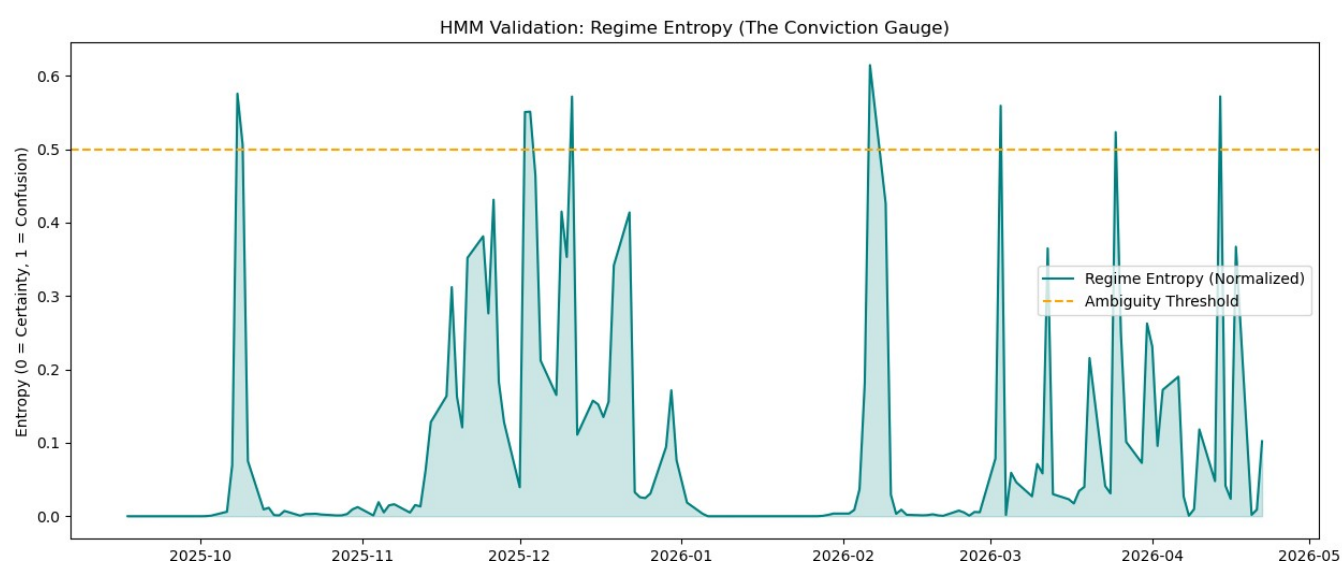


Figure 22. Regime entropy time series. *Source: Author's own work.*

4.8. Robustness Checks

Three robustness checks are reported. First, the rolling AR is recomputed using window lengths of 30, 90, and 120 days. The qualitative pattern is preserved across all window lengths.

Second, MST topology is assessed under bootstrap resampling ($B = 1,000$). The peripheral positioning of Japan Equity is maintained in 94.6% of bootstrap samples. The central positioning of Emerging Markets is maintained in 89.4% of bootstrap samples.

Third, the Bonferroni-corrected Granger result is robust to alternative lag-selection criteria (AIC, BIC, fixed lag 2).

5. Discussion and Limitations

5.1. Robust Findings and Their Risk-Monitoring Implications

Five findings are robust across the methods applied.

First, the Absorption Ratio breaches the 0.50 threshold in all four crisis episodes tested across the 2007–2026 period. The threshold originally identified by Kritzman et al. (2011) on US-equity panels generalises to a 9-asset multi-asset global portfolio across four diverse historical episodes.

Second, the Bonferroni-corrected Granger analysis identifies a single robust empirical regularity: the four major equity blocs collectively Granger-predict Japan Equity at $\alpha^* \approx 0.0007$. This finding converges with the DCC-GARCH end-of-sample topology in which Japan Equity emerges as a high-betweenness node.

Third, the DCC-GARCH baseline confirms persistent cross-asset correlation dynamics with total persistence $a + b = 0.989$ and an implied correlation-shock half-life of approximately 27 trading days.

Fourth, the out-of-sample HMM evaluation confirms that the regime-classification framework correctly identifies the April 2026 episode as a stress regime when calibrated on a 14-year training history.

Fifth, joint F-tests support the characterisation of Oil as a quasi-exogenous asset in conditional mean, while variance decomposition reveals volatility-channel transmission to equities and rates.

5.2. Limitations

Sample brevity of the current-study window. The 261-observation analytical core covers a single 13-month episode.

Granger causality limitations are detailed in Section 3.4: finite-sample size distortions, multiple-testing concerns, heteroskedasticity, non-linear dependence, and parameter instability across sub-periods.

HMM limitations include the Gaussian emission assumption and the challenge of walk-forward evaluation in finite samples.

DCC-GARCH limitations include the scalar specification's assumption that all pairs of assets share the same dynamics parameters.

Tail risk estimation. Non-parametric CVaR and tail dependence estimates at the 5% percentile are based on approximately 13 observations per asset.

Equilibrium interpretation. As stated in Section 2.1, the framework does not derive prices or risk premia from equilibrium theory.

6. Conclusions

This paper presents the Geometry of Risk, an integrated four-layer empirical monitoring framework for multi-asset systemic stress, combining network topology, dynamic causality analysis with multiple-testing correction, tail-risk quantification, and HMM regime classification with out-of-sample evaluation, applied to nine globally representative asset classes.

The principal substantive findings are: (i) the 0.50 Absorption Ratio threshold is breached in all four crisis episodes tested across the 2007–2026 period; (ii) the Bonferroni-corrected Granger analysis identifies a single robust directional finding (four equity blocs jointly Granger-predict Japan Equity at $\alpha^* \approx 0.0007$); (iii) the scalar DCC-GARCH model yields total persistence $a + b = 0.989$ with an implied correlation-shock half-life of approximately 27 trading days; (iv) the long-history-calibrated HMM correctly identifies the April 2026 stress regime out-of-sample; and (v) joint F-tests support a hedged characterisation of Oil as a quasi-exogenous asset transmitting through forecast-error variance rather than conditional mean predictability.

Four extensions define the forward research agenda. First, walk-forward HMM evaluation on samples large enough to support cold-start refits at every step. Second, asymmetric and block DCC specifications. Third, Student-t and copula-based extensions of the tail-dependence analysis. Fourth, transfer-entropy and non-linear Granger extensions.

Abbreviations

The following abbreviations are used in this manuscript:

ADF	Augmented Dickey-Fuller unit-root test
AIC	Akaike Information Criterion
AR	Systemic Absorption Ratio
BIC	Bayesian Information Criterion
CVaR	Conditional Value-at-Risk (Expected Shortfall)
DCC	Dynamic Conditional Correlation
EM	Emerging Markets
FEVD	Forecast Error Variance Decomposition
FRED	Federal Reserve Economic Data
GARCH	Generalised Autoregressive Conditional Heteroskedasticity
GFC	Global Financial Crisis
HMM	Hidden Markov Model
MDS	Multidimensional Scaling
MOVE	Merrill Lynch Option Volatility Estimate (bond volatility index)
MST	Minimum Spanning Tree
OLS	Ordinary Least Squares
OOS	Out-of-Sample
PCA	Principal Component Analysis
RRG	Relative Rotation Graph
SDF	Stochastic Discount Factor
VAR	Vector Autoregression
VaR	Value-at-Risk
VIX	CBOE Volatility Index

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